

REPORT

CATALYSIS

Physical mixing of a catalyst and a hydrophobic polymer promotes CO hydrogenation through dehydration

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In many reactions restricted by water, selective removal of water from the reaction system is critical and usually requires a membrane reactor. We found that a simple physical mixture of hydrophobic poly(divinylbenzene) with cobalt-manganese carbide could modulate a local environment of catalysts for rapidly shipping water product in syngas conversion. We were able to shift the water-sorption equilibrium on the catalyst surface, leading to a greater proportion of free surface that in turn raised the rate of syngas conversion by nearly a factor of 2. The carbon monoxide conversion reached 63.5%, and 71.4% of the hydrocarbon products were light olefins at 250°C, outperforming poly(divinylbenzene)-free catalyst under equivalent reaction conditions. The physically mixed CoMn carbide/poly(divinylbenzene) catalyst was durable in the continuous test for 120 hours.

Selective and rapid removal of water product from a reaction system has been a highly desirable pathway toward boosting catalytic performance in reactions that are restricted by water thermodynamically and/or kinetically (1, 2). Membrane reactors designed to include water-conduction nano-channels could shift the reaction equilibrium (3), but preparation of defect-free membranes at a large scale is challenging. Chemical hydrophobization of the catalyst surface could substantially contribute to reactions by accelerating water diffusion (4–7), but in many cases the chemical interactions might change the structure of the catalyst surface or even block the active sites by hindering access of reactant molecules.

Enabling rapid water diffusion from an unchanged catalyst surface is an attractive alternative. By promoting rapid desorption of water molecules once they are formed on the catalyst surface (2), the sorption equilibrium of water is shifted, as described by $*\text{H}_2\text{O} \rightleftharpoons * + \text{H}_2\text{O}$ (* denotes the catalyst surface sites). We phys-

ically mixed the hydrophobic promoter with the catalyst, unlike previous chemical modifications; in the syngas conversion to olefins with a cobalt-manganese carbide (CoMnC) catalyst, we achieved an increase in light olefin productivity by a factor of up to 3.4 by mixing the catalyst with the promoter. Mechanistic studies revealed that the water molecules rapidly desorbed from the catalyst surface after they formed from CO hydrogenation, which avoided the competitive adsorption with CO reactant, a crucial step in the reaction process.

The CoMnC catalyst was prepared via coprecipitation and carbonization procedures (fig. S1) (8, 9). In the syngas conversion under the given reaction conditions (H_2/CO of 2, 1800 ml $\text{g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, 0.1 MPa, 250°C), the CoMnC catalyst showed a CO conversion of 32.2%, with selectivity for light olefins (C_2^- to C_4^-) of 60.8% (fig. S2 and table S1; the CO_2 product was excluded in calculating the selectivity). In our initial attempt, we physically mixed the CoMnC catalyst with a nonporous poly(divinylbenzene) (PDVB) (water-droplet contact angle 145°, irregular morphology, surface area $<5 \text{ m}^2/\text{g}$; table S2). This hydrophobic polymer has a chemically inert surface (fig. S3) and good thermal stability (10). The mixed catalyst, denoted CoMnC/PDVB, had substantially improved CO conversion (63.5%) and selectivity to light olefins (71.4%) (Fig. 1) relative to the CoMnC catalyst under equivalent reaction conditions. With regard to the C_{5+} by-products over CoMnC/PDVB, the proportions of pentene and hexene were 50.3% and 26.0%, respectively (table S3); these are desired products for the production of high-performance polymers and valuable chemicals (11, 12).

In particular, the molar ratios of olefin to paraffin (*o/p*) in these by-products were very high. For example, the *o/p* ratio of C_6 molecules is 14.7, which is favorable for separation and purification of desired products (13). A higher gas hourly space velocity (GHSV) of 3600 ml $\text{g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$ decreased the CO conversion to 15.4% on the CoMnC catalyst, but the CoMnC/PDVB still showed a remarkably high CO conversion of 50.7% (table S1). Under a GHSV of 7200 ml $\text{g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, the CoMnC catalyst exhibited poor CO conversion of 5.5%, whereas the CoMnC/PDVB still showed CO conversion of 19.4%. In this case, the space-time productivity of light olefins on the CoMnC/PDVB reached as high as 7.1 mmol $\text{g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, which exceeded the rate for CoMnC (2.1 mmol $\text{g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, carbon basis) by a factor of 3.4 (fig. S4). In addition, a CO_2 selectivity of 46.0 to 48.5% was obtained in these cases, similar to that in general Fischer-Tropsch synthesis to olefins (8, 14) and OX-ZEO (oxide-zeolite) reaction processes for converting syngas to olefins (15, 16). Adjusting the H_2/CO ratio to 3.6 and introducing a small amount of CO_2 in the syngas feed reduced the CO_2 selectivity in the products to 23.7%, giving a one-pass yield of light olefins at 28.0% (CO_2 included in calculating the olefin yield; table S4).

We compared the performance of CoMnC/PDVB to different catalysts tested previously in syngas conversion to light olefins. The data in fig. S5 show the comparison of the selectivity-conversion results reported for various catalysts in their stable period during the reaction (table S5). The general Fischer-Tropsch synthesis to olefin processes produced olefins having a wide carbon number distribution within the range $\text{C}_1\text{--C}_{20}$ (17, 18). The selectivity of 53.0% to light olefins, with a CO conversion of 80.0% over Fe-based catalysts, has been reported as one of the most efficient processes (14), requiring a high reaction temperature of 340°C. The OX-ZEO process (15), which combines the cascade reactions of CO hydrogenation over metal oxide and C-C coupling of methanol or ketene intermediates over zeolite, showed superior selectivity to light olefins but yielded low CO conversion (e.g., 17.0% over the $\text{ZnCrO}_x/\text{SAPO-34}$ catalyst) at even higher temperatures (400°C). Relative to these processes, the reaction over the CoMnC/PDVB catalyst proceeded at lower reaction temperatures with high selectivity to light olefins and efficiently suppressed methane formation. The CoMnC/PDVB catalyst also exhibited enhanced catalytic performance relative to the bare CoMn catalyst, which has been regarded as a superior catalyst for low-temperature syngas conversion to light olefins (8).

The catalytic performance of CoMnC/PDVB catalyst was influenced by the manner of mixing of the CoMnC and PDVB components (Fig. 2, A and B, table S6, and fig. S6). Compared with a

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Fig. 1. Catalytic data in light olefin production from syngas. (A and B) CO conversion (A) and hydrocarbon distribution (B) over the CoMnC and CoMnC/PDVB catalysts. Reaction conditions: 1.0 g of CoMnC catalyst or mixture containing 1.0 g of CoMnC and 1.0 g of PDVB, $\text{H}_2/\text{CO}/\text{Ar}$ at 64/32/4, $1800 \text{ ml g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, 0.1 MPa, 250°C . The error bounds were estimated by repeating the experiment more than six times. Inset in (A): Water-droplet contact angles (CAs) of CoMnC and CoMnC/PDVB. *Reaction using a feed gas of $\text{H}_2/\text{CO}/\text{Ar}/\text{CO}_2$ at 68/19/3/10.

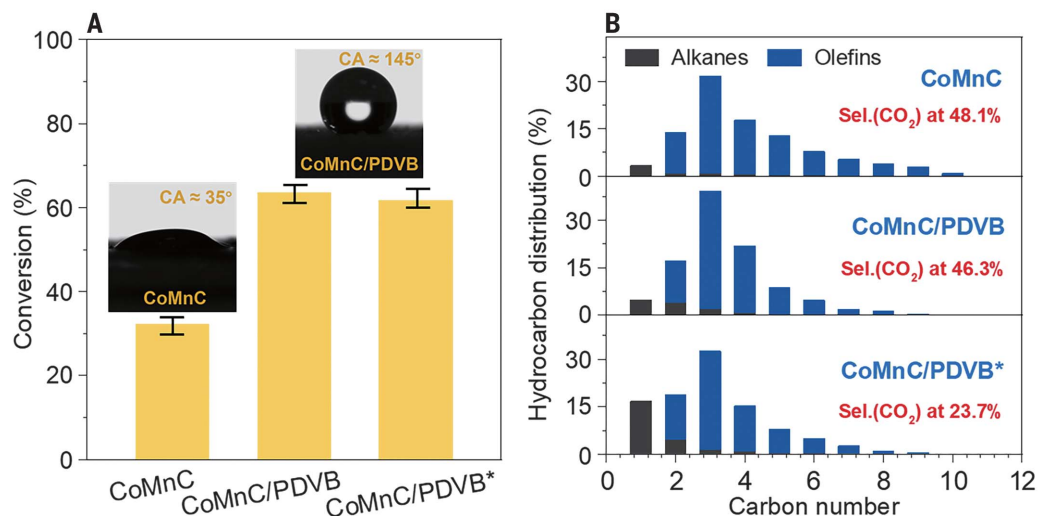
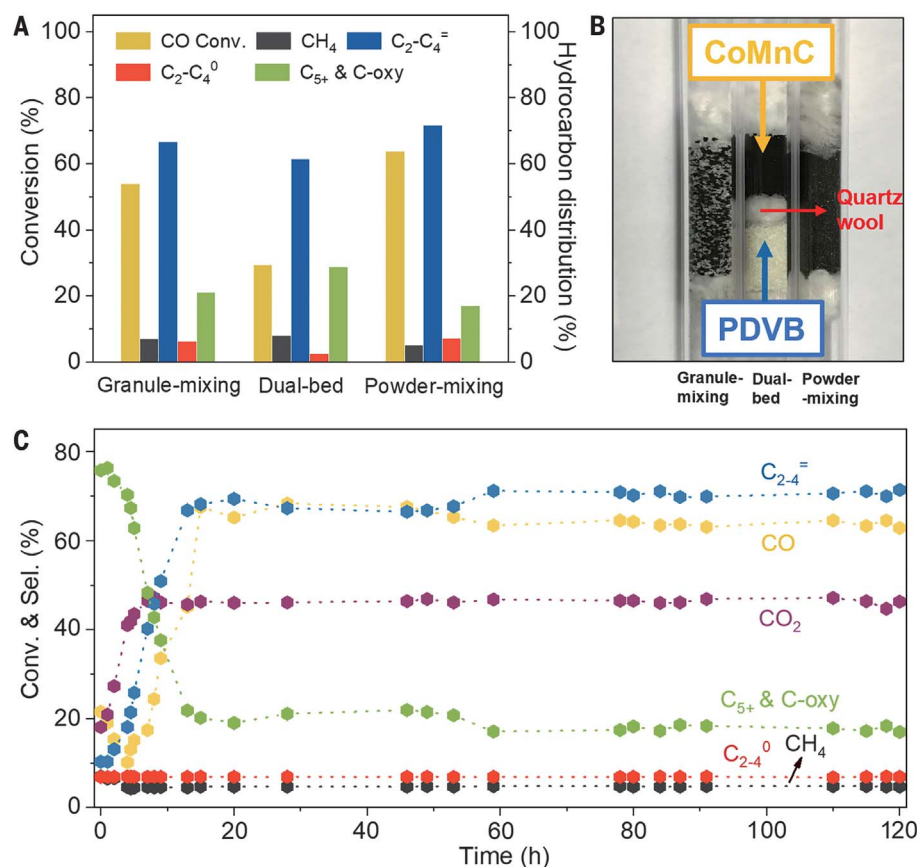


Fig. 2. Effect of different physical mixture methods and catalyst durability. (A) Syngas conversion performance of CoMnC/PDVB produced using different mixing techniques. Reaction conditions: 1.0 g of CoMnC and 1.0 g of PDVB, H_2/CO at 2, $1800 \text{ ml g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, 0.1 MPa, 250°C . (B) Photographs of the catalyst beds with different mixing techniques. (C) Durability of CoMnC/PDVB catalyst in the syngas conversion to light olefins. Reaction conditions: 1.0 g of CoMnC catalyst physically mixed with 1.0 g of PDVB (powder-mixing), H_2/CO at 2, $1800 \text{ ml g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$, 0.1 MPa, 250°C .



powder mixture of the CoMnC and PDVB, the granule mixture, which was prepared by granulating the CoMnC and PDVB components and then mixing them together (40 to 60 mesh), showed lower CO conversion of 53.7% and similar $\text{C}_2\text{-C}_4$ olefin selectivity (66.4%). In a dual-bed reactor, the PDVB was packed below the CoMnC catalyst bed and separated by a layer of inert quartz sand. The CO conversion

of 29.1% and lower olefin selectivity of 61.2% were similar to those of bare CoMnC catalyst. This result indicated that PDVB is inert for the reaction and that the promotion with PDVB required physical mixing with the CoMnC catalyst.

The CoMnC/PDVB catalyst was used in a continuous reaction test to evaluate durability. After activation for ~15 hours, the CO conver-

sion was constant at steady state with an average value of ~64.7% (Fig. 2C). Even after reaction for 120 hours, the CO conversion was well maintained at 62.8% with stable CoMnC and PDVB components, confirming the good durability of the CoMnC/PDVB. In this process, the selectivity of light olefins was also constant at 70.0%, with an average productivity at $5.9 \text{ mmol g}_{\text{CoMnC}}^{-1} \text{ hour}^{-1}$. During the test, the

selectivities for undesired methane and C_2 – C_4 alkanes remained lower than 5.0% and 7.0%, respectively. The selectivity for C_5 – C_8 products was ~18.0%, and 92.0% of those are C_5 – C_8 olefins (19, 20).

To understand the promotion of PDVB in the syngas conversion, we used temperature-programmed surface reaction mass spectrometry (TPSR-MS) by feeding syngas to the catalysts. The CoMnC catalyst has been reported to have superior activity for hydrogen activation and cleavage of the C–O bond (8, 21–23), and these steps can be identified in TPSR-MS tests. The dependences of propylene, methane, and water signals ($m/z = 42$, 16, and 18, respectively) on reaction temperatures (Fig. 3A) show that on the CoMnC catalyst, the water and methane signals appeared at 166° and 203°C, which we assigned to the C–O cleavage and hydrogenation reaction. At 217°C, the propylene signal began to appear because the C–C coupling step occurred after C–O dissociation. Similar signals also appeared on the CoMnC/PDVB catalysts, but the signals of propylene and water were obviously stronger

than those on the bare CoMnC, indicating that the physically mixed PDVB indeed boosted the activity.

It might be expected that the PDVB could participate in the CoMnC carbonization that leads to the distinguishable catalytic performances (22, 24). We excluded this hypothesis by characterizing the CoMnC phase change as a function of reaction time, which showed negligible difference in the x-ray diffraction (XRD) patterns of the CoMnC with and without PDVB during the reaction periods (fig. S7); this result was also supported by the TEM characterization (fig. S8). After removal of the PDVB component from the used CoMnC/PDVB catalyst, the resulting CoMnC component exhibited performance comparable to that of the as-prepared CoMnC (table S1). These results further indicate that the physical mixture with PDVB did not change the catalyst structure, and they are consistent with the high stability of PDVB at the reaction temperature (figs. S9 to S12).

The previous strategies of chemically modifying the catalyst surface with hydrophobic organosilanes were developed to improve water

resistance in syngas conversion (25–27). Following this route, we also modified the CoMnC catalyst using tetraethyl orthosilicate and dimethyl diethyloxysilane, but this resulted in lower CO conversion over the CoMnC@Si and CoMnC@Si-*c* catalysts relative to the bare CoMnC catalyst (table S1). XRD patterns of the used catalysts showed the presence of oxide phases and suggested that hindered carbonization led to formation of the active CoMnC carbides (fig. S13) (27).

We also varied the amount of PDVB in the catalyst bed. PDVB/CoMnC weight ratios of 0.5, 1.0, and 1.5 resulted in distinguishable CO conversions at 56.4%, 63.5%, and 70.5%, respectively (table S7). The sensitivity of CO conversion to PDVB content indicated that it plays a crucial role in catalysis. Although PDVB in the physical mixture did not change the catalyst structure, it might optimize the water diffusion because of its hydrophobicity. Thus, we studied the role of added water in CO conversion over the CoMnC catalyst (Fig. 3B). Considering that the water production rate was 8.8 to 23.5 mg $g_{CoMnC}^{-1} \text{ hour}^{-1}$ in the CoMnC/PDVB-catalyzed syngas conversion (calculated according to the oxygen balance in the reaction system and the amount of collected water product after reaction; fig. S14), we added water at this rate to investigate its influence on the CO conversion. The CoMnC catalyst showed CO conversion of 33.5% at the beginning of the reaction without water injection, which then decreased to ~8.2% (average value) after water injection with a feed rate of ~10.5 mg $g_{CoMnC}^{-1} \text{ hour}^{-1}$. Interestingly, addition of PDVB to the catalyst efficiently minimized the negative effect of water, which exhibited only a relatively slight decrease in CO conversion to ~57.4% under the equivalent water feed. The activity of the CoMnC catalyst was continuously reduced as more water was added to the feed gas (fig. S15), and the CO conversion dropped to 4.7% with water feed rate of ~20.0 mg $g_{CoMnC}^{-1} \text{ hour}^{-1}$. These data confirmed the water-restricted feature of the CoMnC-catalyzed syngas conversion. In contrast, the profile of CO conversion as a function of water concentration was flatter over the CoMnC/PDVB catalyst. This hydrophobic material may have helped the water product to rapidly desorb after it was formed on the CoMnC surface and also hindered its readorption (fig. S16), which would free up active sites for the continuous conversion of more CO molecules.

This hypothesis was supported by a pulse experiment to explore the CO adsorption on the catalyst surface with and without water injection (Fig. 3C). The CoMnC and CoMnC/PDVB catalysts were localized within the flowing He atmosphere at 250°C, with periodic pulsing of the CO or mixture of CO and water. In the test without water, the CO signals on

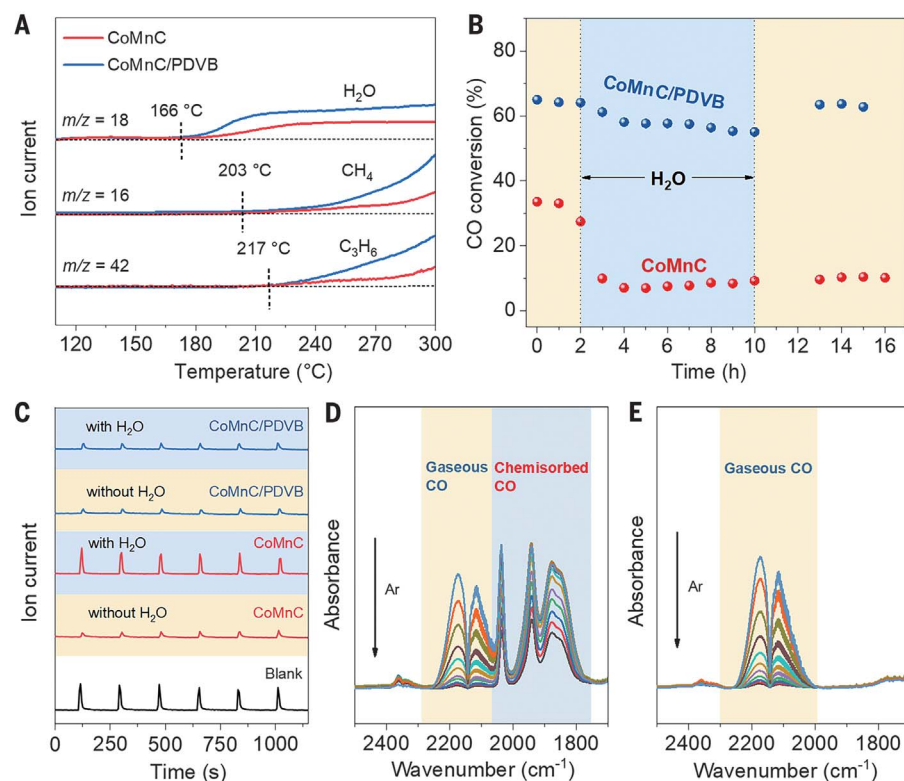


Fig. 3. PDVB-optimized water sorption. (A) The dependences of water, methane, and propylene signals (m/z at 18, 16, and 42) on temperature-programmed surface reaction by feeding syngas to the CoMnC and CoMnC/PDVB catalysts. (B) Data showing the influence of water on syngas conversion with CoMnC and CoMnC/PDVB. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of PDVB (powder-mixing), H₂/CO at 2, 1800 ml $g_{CoMnC}^{-1} \text{ hour}^{-1}$, 0.1 MPa, 250°C. The water feed rate was ~10.5 mg $g_{CoMnC}^{-1} \text{ hour}^{-1}$. (C) Transient response curves obtained during pulses of 10% CO/He (5 ml/min) into pure He flow (30 ml/min) at 250°C over the CoMnC and CoMnC/PDVB catalysts. CO flowed through the water at 40°C for introducing water to the catalysts. (D and E) CO desorption in situ FTIR spectra of anhydrous CoMnC (D) and water-pretreated CoMnC catalysts (E).

both catalysts were extremely weak compared with that in the blank run without catalysts, indicating that efficient CO adsorption occurred on both CoMnC and CoMnC/PDVB catalysts. In these cases, the CO signals were similar, revealing the negligible effect of PDVB on CO sorption, and the CO adsorption dominantly occurred on the CoMnC. When water was fed into the reaction (~ 7 vol% in CO), the CO pulse peaks on the CoMnC catalyst were markedly increased, with the intensities similar to that in the blank run, revealing the hindered CO adsorption by competition with water.

Under the equivalent test, the CO pulse peaks were still weak on the CoMnC/PDVB catalyst, similar to the result of the water-free test, confirming the negligible influence of water on CO sorption in the presence of PDVB. A similar phenomenon was observed in the equivalent tests by changing the water feed amount (fig. S17). On the basis of these results, we conclude that the water on the catalyst surface could hinder the CO adsorption, while the PDVB efficiently shifted the sorption equilibrium described by $\text{*H}_2\text{O} \rightleftharpoons \text{*} + \text{H}_2\text{O}$ by accelerating the rapid water desorption and hindering its readsorption. These features led to a higher proportion of free catalyst surface for continuous conversion of CO, boosting the syngas conversion to olefins (fig. S18).

In situ Fourier transform infrared (FTIR) spectra characterizing the CO hydrogenation on the CoMnC and CoMnC/PDVB catalysts are shown in fig. S19. Introducing CO and hydrogen to the CoMnC catalyst led to the formation of obvious bands at 2717 to 2955 cm^{-1} , 1320 to 1527 cm^{-1} , and $\sim 1039\text{ cm}^{-1}$, which were assigned to the olefin species from CO hydrogenation and C-C coupling (28–30). The broad signals at $\sim 1600\text{ cm}^{-1}$ and 3400 to 3600 cm^{-1} appeared and continuously increased with reaction time because of the water product adsorbed on the catalyst surface (31–34). On the CoMnC/PDVB catalyst, the signals of olefin products were solely observed with an extremely weak water signal, suggesting the rapid desorption of water once it is formed on the catalyst.

The effect of water on CO adsorption was further explored (Fig. 3, D and E). Introducing CO to the CoMnC catalyst led to signals of chemically adsorbed CO on the CoMnC surface (22, 33). When water was co-fed with CO, the signals of chemically adsorbed CO were almost undetectable with only the bands of gaseous CO (34), confirming that CO adsorption was hindered under competition with water. This phenomenon might explain the negative effect of water on the syngas conversion by the competitive adsorption, in good agreement with the results of pulse experiments. In contrast, the catalyst containing PDVB exhibited comparable CO adsorption with and without water injection in a FTIR

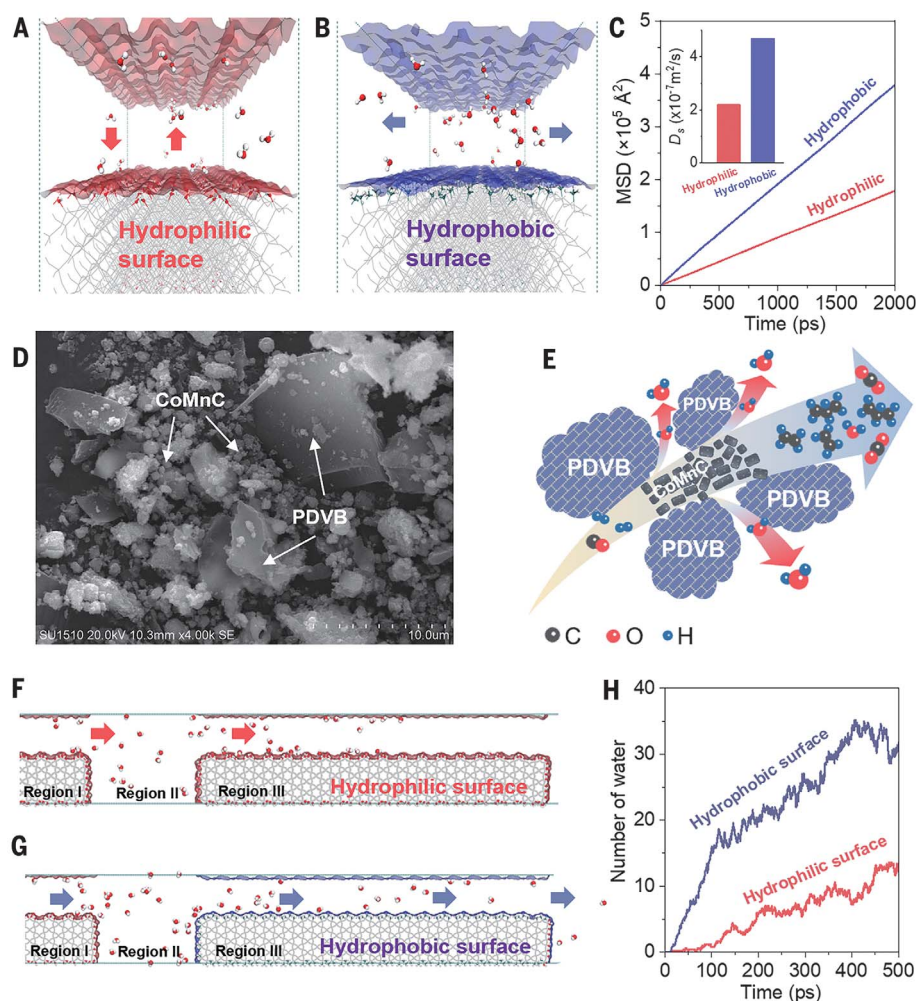


Fig. 4. Theoretical simulation. (A and B) Models showing the water diffusion within regions surrounded by hydrophilic and hydrophobic surfaces. (C) Mean square displacement (MSD) and diffusion coefficient (D_s) showing the water diffusion efficiency at 250°C . (D) SEM image of the CoMnC/PDVB granule. (E) Scheme showing the escape of water from the CoMnC surface through the region surrounded by PDVB. (F and G) Models showing the water escape through different regions. The regions I are hydrophilic and the regions III are hydrophilic and hydrophobic, respectively. (H) The number of water molecules that escaped from region III from an initial state with 100 water molecules on region I as a function of time.

study (figs. S20 and S21). According to previous studies, less water on the catalyst surface might reduce the concentration of carbon-based intermediates and enhance the hydrogenation activity (35, 36), which could explain the improved selectivity of short hydrocarbons and slightly reduced o/p ratios on the CoMnC/PDVB catalyst relative to bare CoMnC.

In addition, the olefin products adsorbing on the catalyst surface might also hinder the CO conversion to some extent, as confirmed by catalysis studies with ethylene in the syngas feed (fig. S22). However, further tests suggested a negligible effect of PDVB on olefin sorption (fig. S23). Thus, PDVB accelerated syngas conversion in the catalytic tests; this was primarily attributed to its hydrophobicity in removing water rather than olefins. We further investigated the influence of promoter wettability

on water diffusion by a theoretical simulation, in which we explored water molecule diffusion in a region surrounded by hydrophilic or hydrophobic surfaces (Fig. 4, A and B). Water diffusion efficiency was quantified by the diffusion coefficient D_s . The hydrophilic surface interacted with water molecules to slow down the transportation (Fig. 4C), giving $D_s = 2.2 \times 10^{-7} \text{ m}^2/\text{s}$. In the region surrounded by hydrophobic surfaces, the water diffusion was accelerated with obviously higher $D_s = 4.7 \times 10^{-7} \text{ m}^2/\text{s}$ (Fig. 4C), given the relatively weak interaction between water and the hydrophobic surface (37). The theoretical simulation therefore gives a qualitative trend of the diffusion of water molecules along the different surface.

In our work, the CoMnC catalyst and PDVB were physically mixed and randomly distributed

in the catalyst granules. As observed in the scanning electron microscope (SEM) image of an actual CoMnC/PDVB catalyst granule (Fig. 4D), the CoMnC and PDVB were packed tightly and disordered, with intergranular distances ranging from nanometers to micrometers (Fig. 4E). We studied how the hydrophobic promoter affects the escape of water that is produced on the relatively hydrophilic surface (e.g., the CoMnC surface). In Fig. 4, F and G, we show models simulating the water escape from the solid surface (region I) through the region surrounded by hydrophobic or hydrophilic surfaces (region III), respectively. Notably, these regions were separated from each other (region II) for simulating the physical mixing with intergranular distances between the CoMnC and PDVB granules in the catalyst (figs. S24 and S25).

The number of escaped water molecules as a function of diffusion time in the systems with hydrophobic and hydrophilic promoters, from an initial state with 100 water molecules on region I (Fig. 4H), indicated that more water molecules escaped through the hydrophobic channel than from the hydrophilic channel under the equivalent conditions. For example, after 500 ps, ~32% of the initial water molecules escaped from the model with the hydrophobic channel, whereas only 13% escaped from the model with the hydrophilic channel. The influence of water concentration on diffusion rate was also simulated by regulating the number of water molecules in region I of the initial state (e.g., 25, 50, and 100 water molecules; fig. S26). The results showed that increasing the concentration of water molecules on the hydrophobic model surface could accelerate the diffusion rate of water molecules, which helps to explain the rapid water diffusion in the syngas conversion reaction with continuously produced water molecules. The models showed that the hydrophobic promoter physically regulated the catalyst by accelerating the water diffusion, in good agreement with the experimental results.

PDVB-promoted water sorption can also be directly observed through a model experiment of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ dehydration, because of its color change from blue to white upon dehydration. After mixing a small amount of PDVB to the $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (0.75 wt% of PDVB in the mixture), the color change was obviously accelerated, as confirmed by the photographs in fig. S27. This result again confirms that PDVB promoted water desorption and hindered readsorption when physically mixed.

We prepared physical mixtures of CoMnC catalyst with different materials whose wettability was distinguishable. When nanopores were introduced to the PDVB (two nanoporous PDVB materials with distinguishable surface areas at 488.2 and 623.3 m^2/g ; table S2 and figs. S28 to S31), the CO conversion further

improved to 88.7% and 92.4% over the CoMnC/nanoporous PDVB catalysts, but selectivity for light olefins was at 53.4% and 37.6% (table S1). Relative to CoMnC/PDVB, the CoMnC/nanoporous PDVB catalysts showed higher selectivity for the heavier olefins (C_5 to C_8 selectivity of 33.7% and 43.7%, respectively). This phenomenon might be the result of the high adsorption capacity of nanoporous PDVB for the olefin products (fig. S32), which prolonged the retention time of olefin products in the catalyst bed to benefit chain growth. These data confirmed that product distribution could be adjusted by changing the nanoporosity of the PDVB promoter.

Similar trends were also observed in the reaction with the CoMnC catalyst mixed with methyl group-modified silica with a hydrophobic surface ($\text{SiO}_2\text{-Me}$, figs. S33 to S35), which showed CO conversion at 74.7% and lower selectivity for $\text{C}_2\text{-C}_4$ olefins at 45.5% (table S1). Higher selectivity for $\text{C}_5\text{-C}_8$ products and lower *o/p* ratios were obtained than with the reaction over CoMnC/PDVB. When graphite, a hydrophobic carbon material, was mixed with CoMnC (CoMnC/Gra, fig. S36), the CO conversion was 53.0% with 67.3% selectivity for light olefins (table S1). Given that the graphite is earth-abundant and extremely cheap, our strategy for shifting water-mediated sorption equilibrium could be implemented simply by mixing hydrophobic graphite with the current catalysts.

In addition, when relatively hydrophilic materials (such as a mixture of PDVB, SiO_2 , and hydrophilic polymers; figs. S37 to S40) were used in the CoMnC-catalyzed syngas conversion, the CO conversion was markedly reduced (tables S8 to S10). For example, polystyrene (PS, fig. S40), which has composition similar to PDVB but is more hydrophilic, failed to promote the CoMnC-catalyzed syngas conversion, showing CO conversion at 10.7% with methane selectivity of 37.8% and $\text{C}_2\text{-C}_4$ olefin selectivity of 37.8% (table S10). These data show the importance of a hydrophobic promoter.

Our approach could be used to upgrade industrially catalytic processes without modifying the catalysts themselves. In addition, the strategy is conceptually different from catalyst hydrophilization with organosilanes, described as a chemical modification route, where the conversion was not obviously improved in syngas conversion but the CO_2 selectivity was reduced by hindering the undesired water-gas shift (4). This difference might result from the distinguishable distances between the active site and the hydrophobic surface for these different systems. Considering that many hydrogenation reactions are strongly affected by water, the physical regulation method using a promoter with desired wettability could guide the design of more efficient catalysts in the future.

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ACKNOWLEDGMENTS

We thank F. Chen for help in SEM characterization. **Funding:** Supported by the National Key Research and Development Program of China (2021YFA1500404), National Natural Science Foundation of China (U21B20101, 21932006, 22032005, 22102143, and 22125304), China Postdoctoral Science Foundation (2021M702803), and National Postdoctoral Program for Innovative Talents (BX20200291). **Author contributions:** W.F. and C.W. performed the catalyst preparation, characterization, catalytic tests, and data analysis. Z.Q.L. and A.Z. performed the theoretical calculation and wrote the corresponding part. L.L., H.L., X.Q., S.X., and L.J.L. participated in the catalyst characterization and discussion. L.W. and F.-S.X. designed the study, analyzed the data, and wrote the paper.

Competing interests: The authors declare that there is no conflict of interest. **Data and materials availability:** All data are available in the manuscript or the supplementary materials. **License information:** Copyright © 2022 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.abo0356

Materials and Methods

Figs. S1 to S40

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Submitted 9 January 2022; resubmitted 18 April 2022

Accepted 24 May 2022

10.1126/science.abo0356



Supplementary Materials for

Physical mixing of a catalyst and a hydrophobic polymer promotes CO hydrogenation through dehydration

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Science **377**, 406 (2022)
DOI: 10.1126/science.abo0356

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Materials

Tetraethylorthosilicate (TEOS, >99%), divinylbenzene (DVB), azobis isobutyronitrile (AIBN), *n*-hexane (>99%, GC), polystyrene (PS) and SiO₂ (AR) were obtained from Aladdin Chemical Reagent Company. Diethoxydimethylsilane (DEMS, 99%), ethanol (AR, >99.5%), methanol (AR, >99.5%), Co(NO₃)₂·6H₂O, 50 wt% Mn(NO₃)₂, CuSO₄·5H₂O (99%), ethyl acetate (EA), dimethyl formamide (DMF), and Na₂CO₃ were obtained from Sinopharm Chemical Reagent Co. Ltd. Graphite (Gra) and sodium styrene sulfonate (SSS) were purchased from Shanghai Macklin Biochemical Co. Ltd. Polytetrafluoroethylene (PTFE) was purchased from Adamas Reagent Ltd. Feed gas (containing pure CO, pure H₂, CO/Ar, H₂/CO/Ar, H₂/Ar, C₂H₄, C₃H₆, and CO₂ etc.) were supplied by Hangzhou Jingong special gas Co. Ltd.

Catalyst preparation

Synthesis of CoMnC. The CoMnC catalyst was prepared by a co-precipitation method described previously (8). Typically, 29.4 g of Co(NO₃)₂·6H₂O and 17.9 g of 50 wt% Mn(NO₃)₂ were dissolved in 75 mL of deionized water to obtain solution A (mole ratio of Co/Mn at 2/1). Meanwhile, 24.0 g of Na₂CO₃ were dissolved in 75 mL of deionized water to obtain solution B. Then solutions A and B were simultaneously dropwise added into a beaker containing 200 mL of deionized water under vigorous stirring at 30 °C. In this process, the pH value was maintained at ~8. The obtained suspension was kept with stirring for 1 h and aged for 2 h at 30 °C. After the precipitate was centrifuged, washed with a large amount of deionized water, dried at 100 °C for 10 h, and calcined at 330 °C for 3 h, the CoMn oxide was obtained. By the pre-reducing treatment and catalyzing syngas conversion, the CoMn oxide could be transferred into CoMn carbide (CoMnC).

Synthesis of nonporous PDVB. 0.5 g of AIBN was added into 10 g of divinylbenzene with stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 24 h. The obtained solid was washed with methanol and dried at 100 °C overnight to obtain the nonporous polydivinylbenzene, which was abbreviated as PDVB.

Synthesis of nanoporous PDVB. 10 g of divinylbenzene and 0.5 g of AIBN was dissolved in 50 g of solvent under stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 12 h. The obtained solid was washed with methanol and dried at 100 °C overnight to obtain the nanoporous polydivinylbenzene. In this process, the DMF and EA were used as solvents, and the obtained samples were denoted as PDVB-DMF and PDVB-EA, respectively. In this manuscript, if there was no special notification, the PDVB means the nonporous sample, while the PDVB-DMF and PDVB-EA were denoted as the nanoporous samples.

Synthesis of PDVB-SSS. As a typical run for synthesis of PDVB-SSS with a molar ratio of SSS/DVB at 0.03, 0.5 g of AIBN was added into 9.5 g of DVB and 0.5 g of SSS with stirring at room temperature for 1 h, then the liquor was transferred into an autoclave and thermally treated at 100 °C for 24 h. The PDVB-SSS was obtained by washing with methanol/ethanol and drying at 100 °C overnight. For comparison, the PDVB-SSS materials with molar ratios of SSS to DVB at 0.07, 0.16, 0.27, 0.42 were synthesized from similar procedures except for using 9.0 g of DVB and 1.0 g SSS, 8.0 g of DVB and 2.0 g SSS, 7.0 g of DVB and 3.0 g SSS, 6.0 g of DVB and 4.0 g SSS in the starting mixture, respectively. Note: the SSS could not be completely dissolved in the starting DVB liquor, but this method can form a relatively homogeneous mixture of hydrophobic and hydrophilic individuals in the final materials for varying the wettability.

Synthesis of SiO₂-Me. 7.49 g of TEOS and 0.592 g of DEMS were dropped into a mixed liquor containing 80 mL of ethanol, 100 mL of water, and 6 mL of aqueous ammonia. After stirring at room temperature for 12 h, the liquor was removed by distillation under vacuum to get the solid powder, which was then dried at 100 °C overnight to obtain the amorphous silica modified with methyl groups (SiO₂-Me).

Synthesis of CoMn@Si and CoMn@Si-c. The CoMn@Si was prepared by the modified Stöber method (4). Typically, 3 g of the prepared CoMn oxide was dispersed into 300 mL of ethanol by ultrasonication. Then, 15 mL of TEOS was dropped into the mixture with stirring at room temperature for 4 h, followed by adding 30 mL of ammonia and 120 mL of water. After stirring the mixture at room temperature for another 4 h, the solid product was filtrated, washed with a large amount of ethanol, and dried at 100 °C overnight. Finally, the CoMn@Si sample was obtained. The hydrophobic encapsulation of CoMn@Si was achieved by further modification using organosilane. Typically, 1.5 g of CoMn@Si was pre-dried at 150 °C for 12 h and dispersed into 10 mL of *n*-hexane under stirring at room temperature. Then, 3.0 g of DEMS was continuously dropped into the mixture for 1 h under stirring, followed by an ultrasonic treatment for 3 h. The sample donated as CoMn@Si-c was obtained by filtrating, washing with *n*-hexane, and drying at 100 °C overnight.

Catalyst characterization

Powder X-ray diffraction (XRD) patterns were collected on a Rigaku D/MAX 2550 diffractometer with CuK α radiation ($\lambda = 1.5418 \text{ \AA}$). Nitrogen sorption isotherms were measured using a Micromeritics ASAP 2020M system. Transmission electron microscopy (TEM) images were recorded on a JEM-2100F electron microscope (JEOL, Japan) with an acceleration voltage of 200 kV. Scanning electron microscopy (SEM) experiments were performed with Hitachi SU-8010 electron microscopes. Metal loadings and Si/Al ratios of the catalysts were determined by inductively coupled plasma analysis (ICP, Perkin-Elmer 3300DV). IR characterization was

performed with Thermo Nicolet NEXUS 470 FT-IR system. Thermogravimetric curves (TG) were performed on a SDT Q600 Simultaneous DSC-TGA in flowing air with a heating rate of 10 °C/min. TG-MS profiles were obtained from a METTLER TGA/DSC instrument connected to a mass spectrometer (LCD200M, TILON). ¹³C NMR spectra were obtained from a Bruker Avance III HD 400 MHz spectrometer. Water-droplet contact angles were performed with Belsorp MaX II system. Water adsorption tests were performed with video-based contact angle measuring device (OCA 20, Dataphysics Co. Ltd).

CO-hydrogenation *in-situ* FTIR spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and a high-temperature reaction chamber. The samples were placed into crucible and pre-treated in pure Ar flow at 400 °C for 1 h (flow rate at 30 mL/min) and then the temperature was cooled down to 250 °C, the background data was collected at a stable state. Then, feed gases of CO (flow rate at 10 mL/min) and H₂ (flow rate at 20 mL/min) were simultaneously introduced into the crucible and spectra were recorded at intervals of one minute. After 10 minutes, the CO flow and H₂ flow were switched off, pure Ar flow (flow rate at 30 mL/min) was introduced into the crucible for 2 min, then the spectrum was recorded.

CO-desorption *in-situ* FTIR spectra were recorded using a Nicolet is50 FT-IR spectrometer equipped with an MCT/A detector and a high-temperature reaction chamber. As a typical run, the sample was placed into crucible and pre-reduced in pure H₂ flow at 300 °C for 5 h (flow rate at 30 mL/min), then the temperature was cooled down to 250 °C in pure Ar flow (flow rate at 30 mL/min) to collect the background spectra at a stable state. Subsequently, feed gases of CO (flow rate at 10 mL/min) and H₂ (flow rate at 20 mL/min) were simultaneously introduced into the crucible for an *in-situ* carbonization. After that, the gas was switched to Ar flow for a while, and CO adsorption was performed by switching the Ar flow to a dry CO flow (flow rate at 30 mL/min) at 250 °C for 0.5 h. After the CO adsorption, the system was purged with Ar (flow rate at 10 mL/min) until the gas phase CO was removed. For studying the hindered CO adsorption by water on the catalyst surface, the FTIR study was followed by similar procedures mentioned above except using the humid CO feed, which was realized by water bubbling at 20 °C.

CO adsorption in pulsed experiment was performed using a fixed-bed glass reactor connected to a mass spectrometer (SRD200M, TILON). As a typical run, 0.2 g of CoMnC (or the power mixture of 0.2 g of CoMnC and 0.2 g of PDVB) was localized in the middle of the reactor tube (40-60 mesh). The temperature of catalyst bed was raised to 250 °C in a pure He flow (30 mL/min) and maintain at 250 °C for 1 h. After that, a slight amount of CO (1 mL of mixed gas containing 10% CO/He) was pulsed into the system, the signals of CO were collected by a mass detector. The pulsed experiment was repeated for multiple times (time interval of

CO pulse was ~3 min) in each test. The CO adsorption in pulsed experiment with water injection was performed *via* the similar procedures except using 10% CO/He flow with water bubbling. The CO adsorption in olefin atmosphere was performed *via* the similar procedures except using 10% C₃H₆/He was flow gas instead of pure He.

Propylene breakthrough experiments were performed using a stainless-steel column (4.6 mm inner diameter × 50 mm). The column packed with sample was firstly purged with N₂ flow (30 mL/min) for 24 h at room temperature. After that, 10% C₃H₆/N₂ was introduced with flow rate at 10 mL/min. The propylene concentration in outlet gas was monitored by a gas chromatography (FuLi-9790, FID) equipped with a capillary PLOT-Al₂O₃ column (50 m × 0.53 mm × 0.25 μm).

Temperature-programmed surface reaction (TPSR) was performed using a fixed-bed glass reactor connected to a mass spectrometer (SRD200M, TILON). 0.2 g of CoMnC (or the power mixture of 0.2 g of steady state CoMnC and 0.2 g of PDVB) was localized in the middle of the reactor tube (40-60 mesh) and pre-treated in pure He flow at 150 °C for 1 h (flow rate at 20 mL/min). Then, the temperature was cooled to 45 °C. TPSR profiles (*m/z* at 16, 18, and 42) were collected in flowing syngas (5 mL/min, H₂/CO molar rate at 2) by raising the temperature from 45 to 300 °C with a ramping rate at 10 °C/min. The thermal stability of PDVB at the reaction temperature was evaluated using a chemisorption instrument (iChem700, PhysiChem Instruments Ltd.) connected to a mass spectrometer (SRD200M, TILON).

Catalyst performance

The syngas conversion was performed in a stainless steel fixed-bed flow reactor with an inner diameter of 10 mm. Typically, 2.0 g of composited catalyst (40-60 mesh) with weight ratio of CoMn/promoter at 1 was mixed and localized in the middle of the reactor unless otherwise stated. Additional quartz sands were placed into the reactor to maintain the catalyst bed. Prior to reaction tests, the catalyst was *in-situ* reduced at 300 °C for 5 h in a pure H₂ flow (flow rate at 20 mL/min) at atmospheric pressure. After reduction, the reactor was cooled to 250 °C, and the feed gas (H₂/CO molar ratio at 2) was introduced into the reactor under a pressure of 0.1 MPa. The gaseous products from the reactor were analyzed with two online gas chromatographs (FuLi-9790 gas chromatograph). One is equipped with a TDX-01 column (3 m × 3 mm) and a thermal conductivity detector (TCD), and the other is equipped with a capillary PLOT-Al₂O₃ column (50 m × 0.53 mm × 0.25 μm) and a flame ionization detector (FID). Argon was used as an internal standard for calculating CO conversion and product selectivities (CO₂ and CH₄) in TCD. CH₄ was used as an internal standard for calculating the product selectivities in online FID. The conversions and product selectivities were calculated according to the number

of carbon atoms. The GHSV was given on the basis of the weight of CoMnC in the reactor ($\text{mL} \cdot \text{h}^{-1} \cdot \text{g}_{\text{CoMnC}}^{-1}$), and the promoter was not included.

Computational Methods

Computational Models. Both periodic model and surface model were used for studying the diffusion behaviors of water molecules in simplified hydrophilic and hydrophobic system. For periodic models, the simulation box was $45.5 \times 59.0 \times 32.4 \text{ \AA}$, and 50 H_2O molecules adsorption in the $20 \text{ \AA}$ vacuum layer parallels to the surface (Figs. 4A and 4B). While for surface models, the simulation box was $540.4 \times 39.7 \times 32.4 \text{ \AA}$, and 100 H_2O molecules adsorption in $20 \text{ \AA}$ vacuum layer parallels to the surface. The $-\text{OH}$ and $-\text{CH}_3$ groups were added to SiO_2 (quartz) to represent hydrophilic and hydrophobic surface, respectively. The influence of water concentration on diffusion rate was also simulated by regulating the number of water molecules adsorbed on the model surface (e.g. water molecules at 25, 50, and 100, donated at 25 H_2O model, 50 H_2O model, and 100 H_2O model).

Molecular dynamics (MD) simulation. MD simulations were performed in the canonical ensemble (NVT), where the number of particles (N), volume (V), and temperature (T) were kept constant. The simulated temperature was 250°C and controlled by a Nosé-Hoover thermostat with a coupling time constant of 0.1 ps . The leapfrog Verlet algorithm was used to integrate the Newton's equations of motion with a time step of 0.5 fs . 1.0×10^6 (large surface model) to 1×10^7 (periodic model) time steps were production, and the trajectories were recorded every 1000 steps for studying the diffusion behaviors of water molecules. The COMPASS (38-41) force field was used and the cutoff radius was $12.5 \text{ \AA}$. During the simulation, the Si atoms were fixed and 5 independent MD simulations were carried out for better statistics. All simulations were performed in Computing Center of Innovation Academy for Precision Measurement Science and Technology, CAS and Shenzhen Cloud Computing Center.

Mean square displacement (MSD) and diffusion coefficient (D_s). The MSD of water was defined as equation in the following:

$$MSD(\tau) = \frac{1}{N_m} \sum_i^{N_m} \frac{1}{N_\tau} \sum_{t_0}^{N_\tau} \left[r_i(t_0 + \tau) - r_i(t_0) \right]^2 \quad (1)$$

where N_m was the number of gas molecules, N_τ was the number of time origins used in calculating the average, and r_i was the coordinate of the i -th molecule. In addition, the slope of the MSD as a function of time determined the self-diffusion coefficient, D_s , which was defined according to the so-called Einstein relation.

$$MSD(\tau) = 2nD_s\tau + b \quad (2)$$

where n was the dimension of materials. In this work, $n = 2$ for the diffusion along with the surface. The reported MSD and D_s values were calculated as the average of 5 independent MD trajectories.

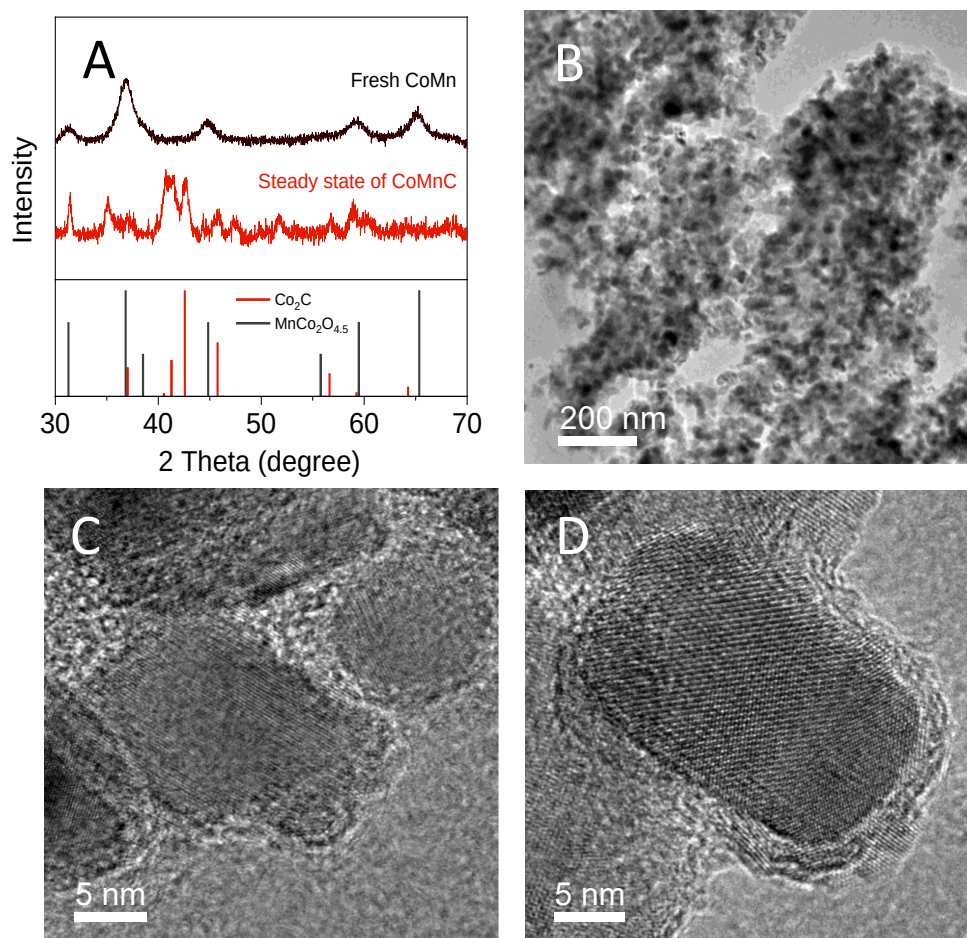


Figure S1. (A) XRD patterns of the as-synthesized CoMn oxide and steady-state CoMnC after catalyzing syngas conversion for 20 h. (B-D) TEM images of the CoMnC catalyst. Inset in (A) showed the standard XRD patterns of the Co_2C and $\text{MnCo}_2\text{O}_{4.5}$ according to PDF # 65-1457 and PDF # 32-0297, respectively.

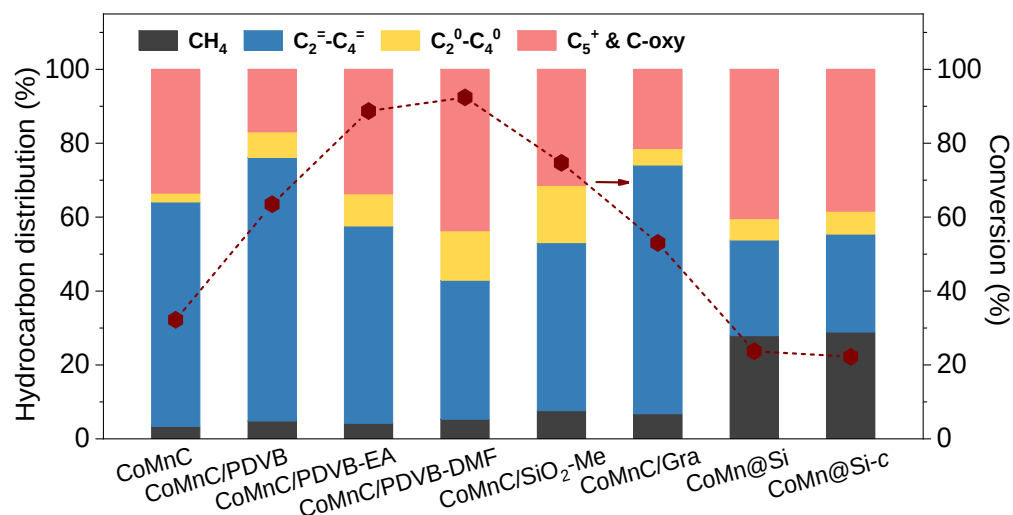


Figure S2. Catalytic data of various catalysts in syngas conversion. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of DPVB, or 1.0 g of silica modified CoMn, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoter. C-oxy means the carbon-containing oxygenates.

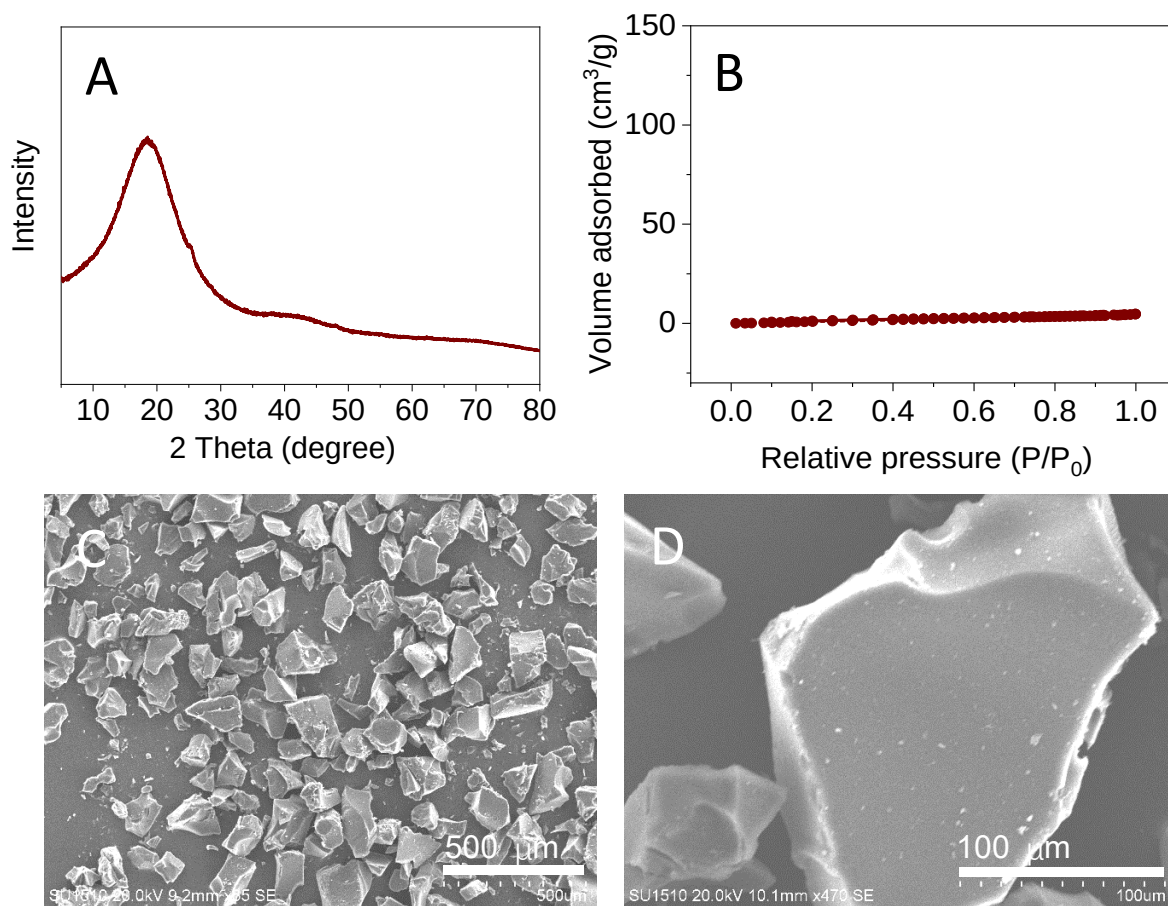


Figure S3. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the PDVB.

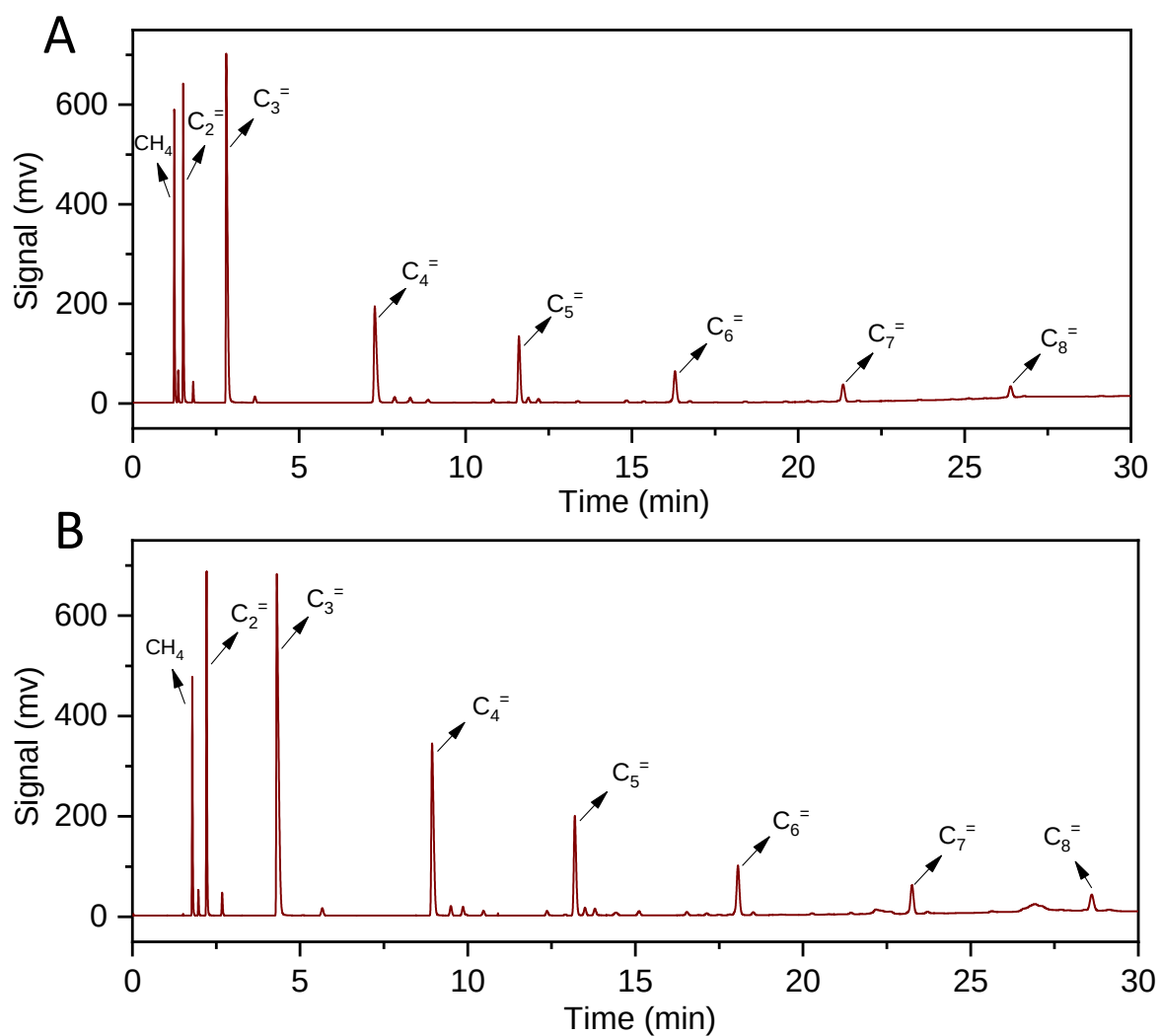


Figure S4. GC curves analyzing the gaseous products (online analysis) in the reaction over the (A) CoMnC and (B) CoMnC/PDVB catalysts.

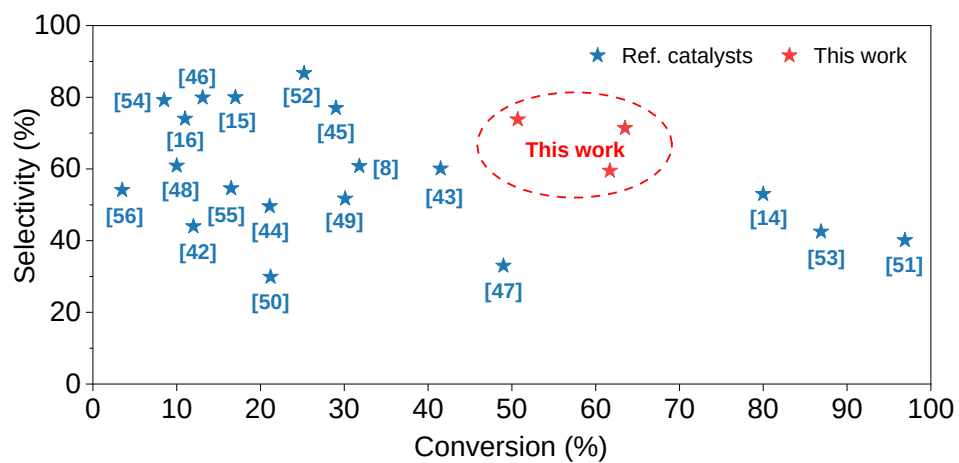


Figure S5. Selectivity of light olefins in the hydrocarbon products as a function of CO conversion over efficient catalysts in literatures and in this work. The CO₂ by-product was not included in the light olefin selectivity.

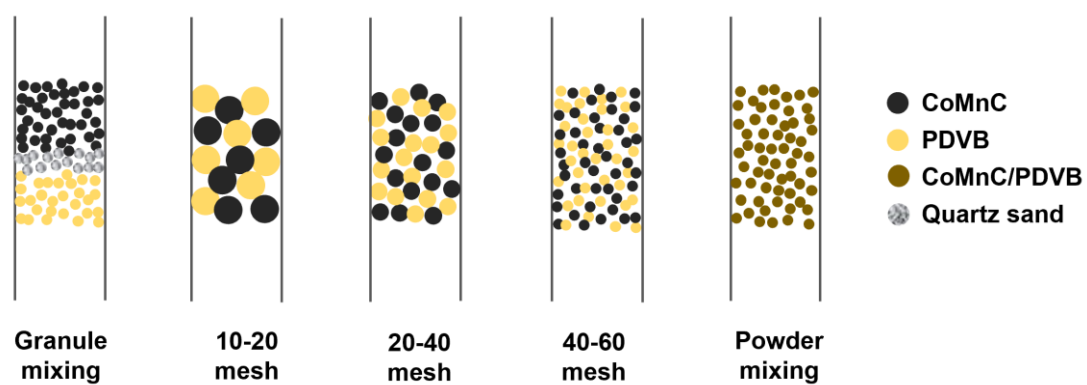


Figure S6. Modes of the CoMnC/PDVB catalysts with different proximities between CoMnC and PDVB.

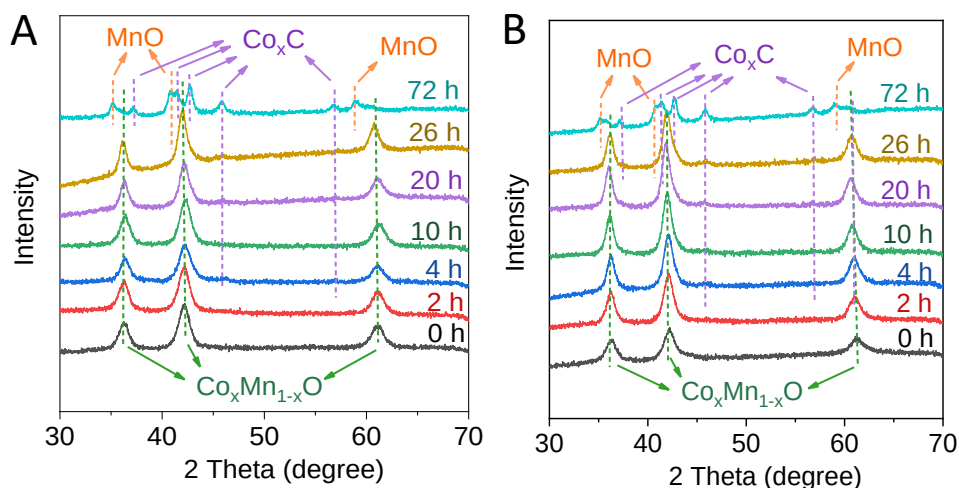


Figure S7. XRD patterns of the (A) CoMnC and (B) CoMnC/PDVB catalysts at different reaction periods. Reaction conditions: 0.2 g of CoMnC or 0.2 g of CoMnC physically mixed with 0.2 g of PDVB, H_2/CO at 2, $1800\text{ mL}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, 0.1 MPa, $250\text{ }^\circ\text{C}$, powder-mixing for CoMnC and PDVB.

Note: In the initial 15 h of the syngas conversion (Fig. 2C), the CO conversion was continuously increased from 10.2% to 67.6% in this period and became constant after 15 h. The selectivity to C_2 - C_4 olefins was also increased from 18.1% to 68.2%, accompanied with the decrease of C_{5+} product selectivity. Such reaction trend is very similar to that in the syngas conversion on general CoMnC catalyst without PDVB, where the conversion/selectivity changed with the CoMnC carbonization in the initial reaction (8).

In addition, this hypothesis on PDVB (or the depolymerized species) participating in the CoMnC carbonization could be excluded by characterizing the CoMnC phase change as a function of reaction time. Negligible difference was observed in the XRD patterns of the CoMnC with and without PDVB during the initial reaction periods (fig. S7), as well as the TEM characterizations (fig. S8). These data further support that the PDVB in physically mixed manner does not change the catalyst structure.

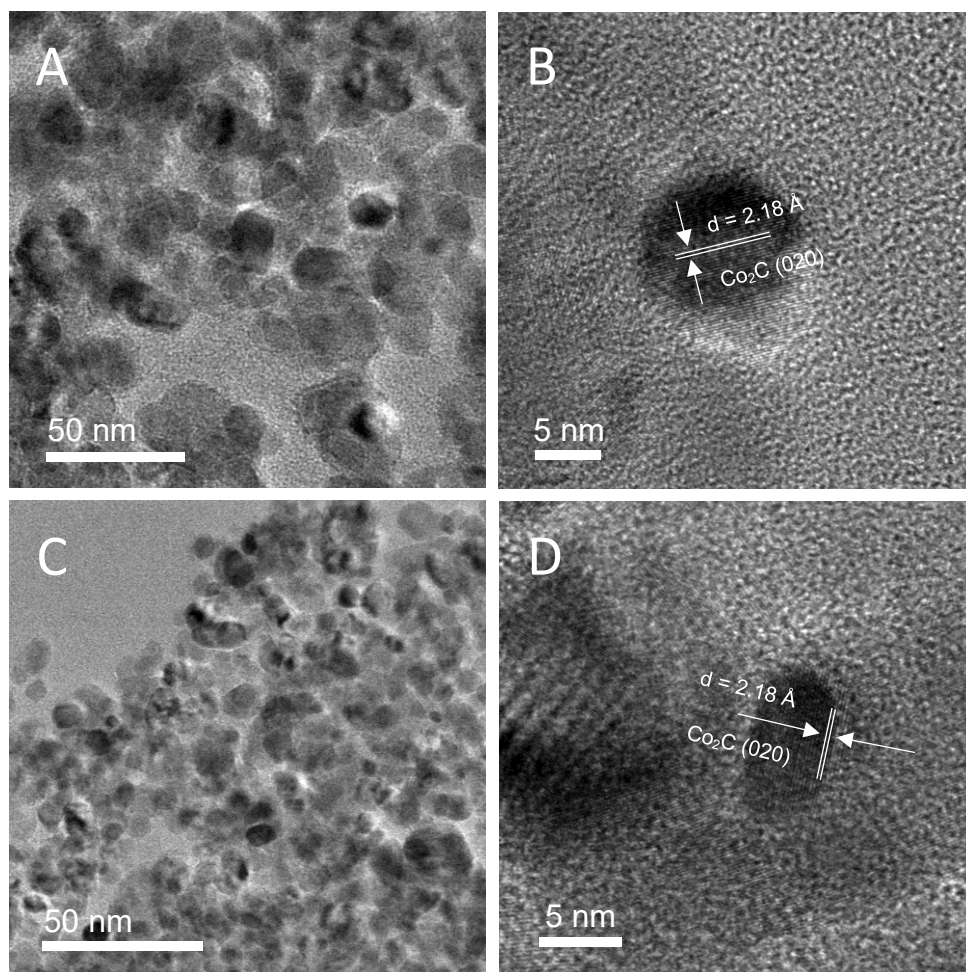


Figure S8. TEM images of the (A,B) CoMnC catalyst and (C,D) CoMnC separated from CoMnC/PDVB catalyst after catalyzing the syngas conversion for 72 h.

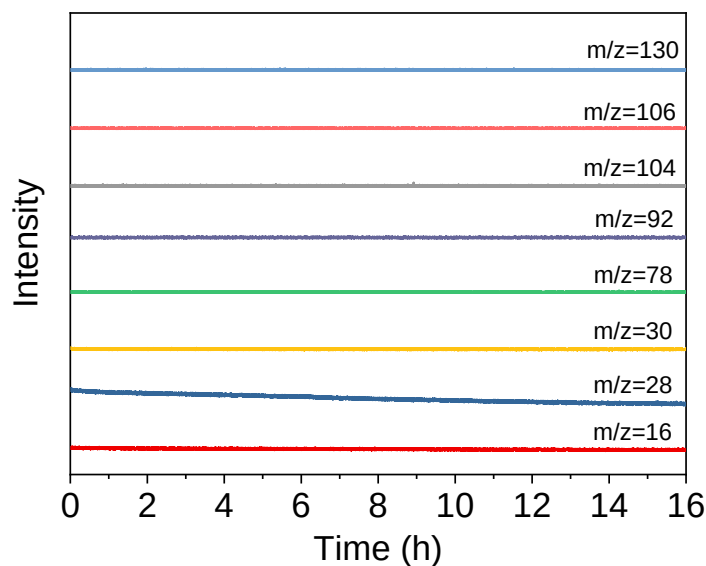


Figure S9. Mass spectra characterizing PDVB stability at 250 °C for more than 16 h.

Note: To evaluate the stability of PDVB, the PDVB was heated in a nitrogen flow at 250 °C and the emission gas was analyzed by a mass spectroscopy. As shown in fig. S8, the signals of possible species from PDVB decomposition (m/z at 16, 28, 30, 78, 92, 104, 106, and 130) were undetectable in the mass spectra. The ^{13}C NMR spectra of PDVB were unchanged before and after the thermal treatment at 250 °C for 10 h (fig. S10). Fig. S11A shows the TG-MS profiles of PDVB, where the weight loss starts at ~ 370 °C with the formation of multiple hydrocarbon species detected by mass spectrometer, suggesting its decomposition at the temperature higher than 370 °C. At 250 °C, the reaction temperature for syngas conversion, the weight loss and decomposed species were completely undetectable. These data confirm the good stability of PDVB under hours of annealing at the reaction temperature. Fig. S11B showed the TG-DSC profiles of PDVB, where the endothermal signal started at about 300 °C, which suggest that the PDVB might melt at temperatures higher than 300 °C but very stable at 250 °C. In addition, we also heated the PDVB granules in a quartz tube at 250 °C for a long period of 50 h, it was observed that the morphology of the granules was unchanged, suggesting the excellent physical stability (fig. S12). This feature might help to exclude the possibility of PDVB melting and coating on the CoMnC surface to change the performances.

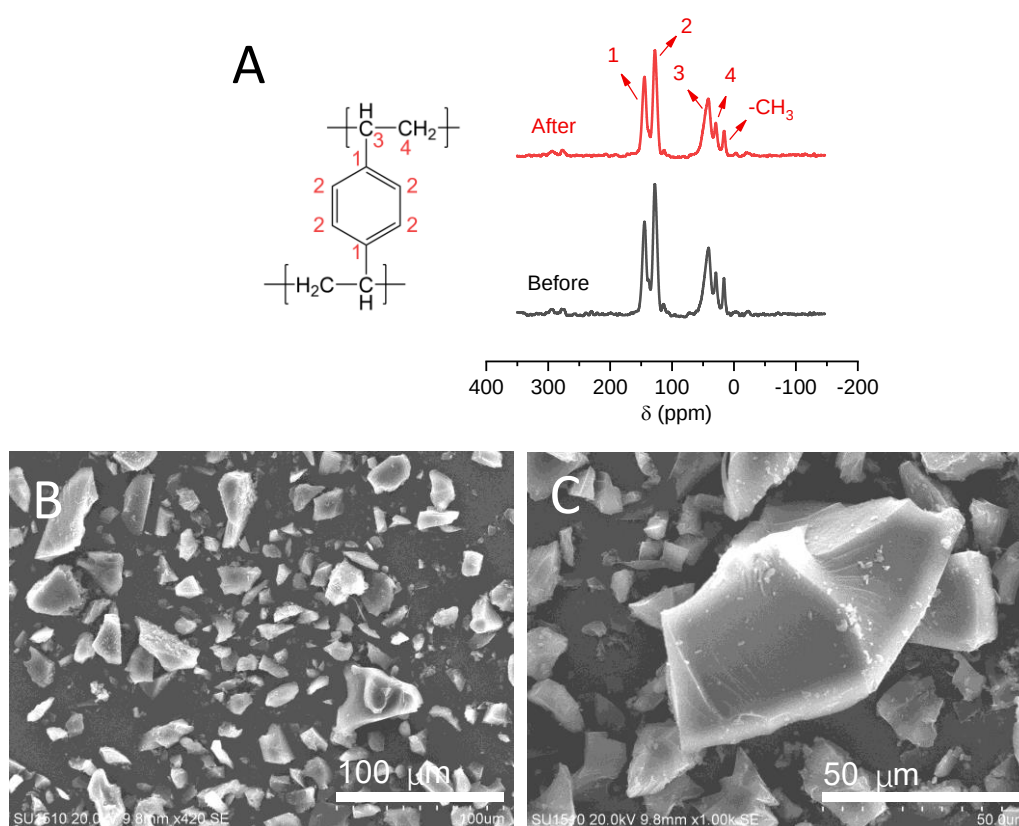


Figure S10. (A) ^{13}C NMR spectra of PDVB before and after treatment at 250 °C for 10 h. (B,C) SEM images of PDVB separated from CoMnC/PDVB catalyst after reaction.

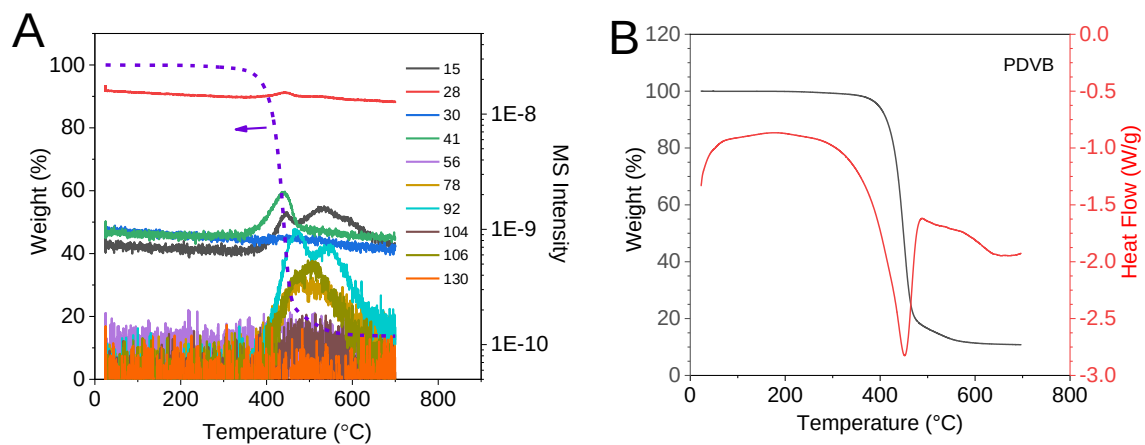


Figure S11. (A) TG-MS and (B) TG-DSC profiles of the PDVB.

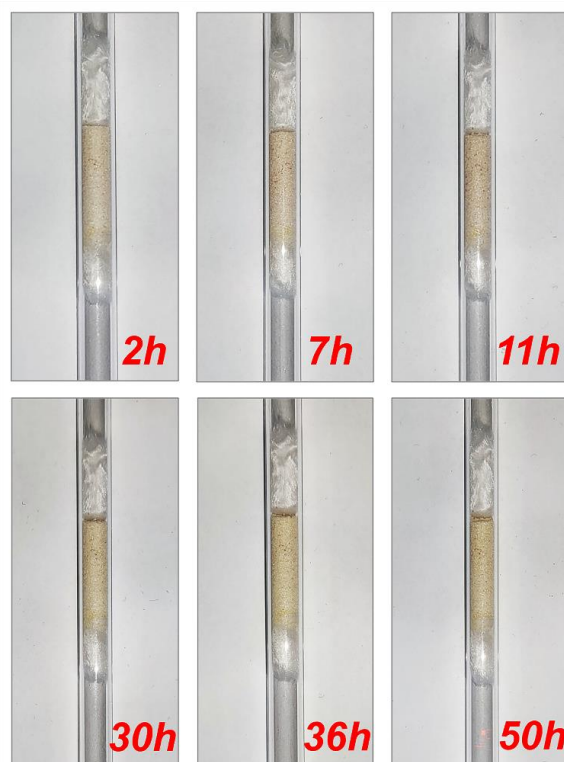


Figure S12. Photographs showing the PDVB granules in a quartz tube with thermal treatment at 250 °C for different periods.

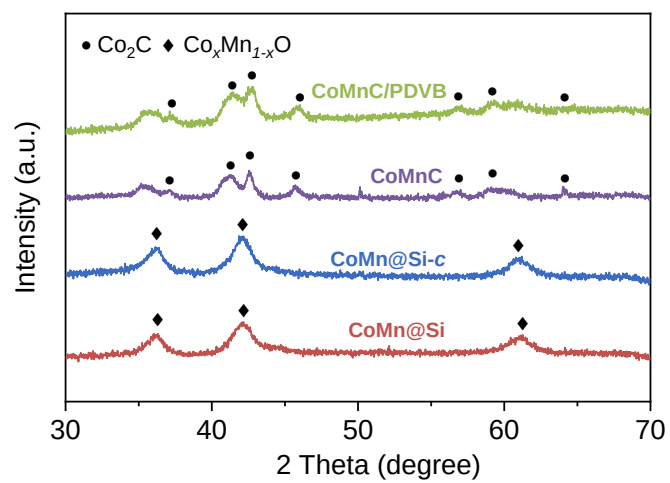


Figure S13. XRD patterns of the used CoMn@Si, CoMn@Si-c, CoMnC, and CoMnC/PDVB catalysts.

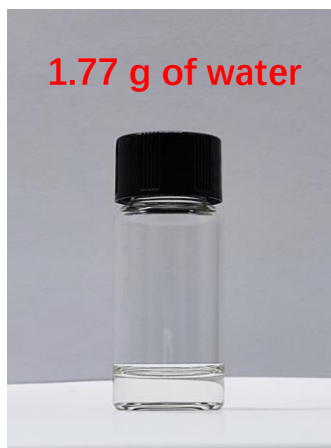


Figure S14. Photograph showing water in the final products from CoMnC/PDVB catalyzed syngas conversion. This water was collected for 102 h.

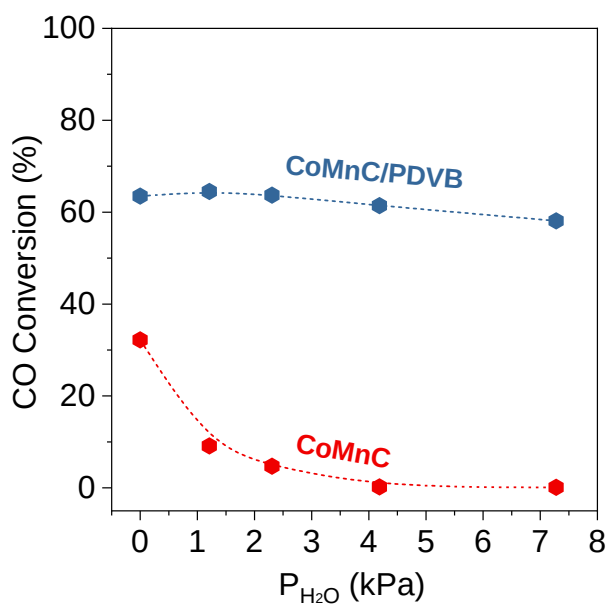


Figure S15. Dependences of CO conversions on water concentrations over the CoMnC and CoMnC/PDVB catalysts. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, H_2/CO at 2, $1800 \text{ mL} \cdot \text{h}^{-1} \cdot \text{g}^{-1}$, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB. Water bubbling with CO flow at 10, 20, 30, and 40 °C to adjust the partial pressures. The water feed rate at $\sim 10.5 \text{ mg g}_{CoMnC}^{-1} \text{ h}^{-1}$ was realized by water bubbling at 10 °C, corresponding to the partial pressure of $\sim 1.2 \text{ kPa}$.

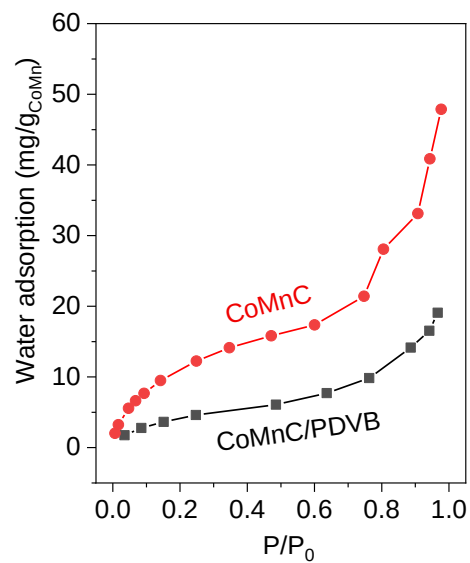


Figure S16. Water adsorption tests on the CoMnC and CoMnC/PDVB.

Note: We tested the water adsorption ability of the CoMnC and CoMnC/PDVB at different relative humidity. The CoMnC always showed higher water adsorption ability than CoMnC/PDVB at the equivalent relative pressures of water (P/P_0). These data confirm the efficiently hindered water adsorption on the CoMnC by mixing with hydrophobic PDVB.

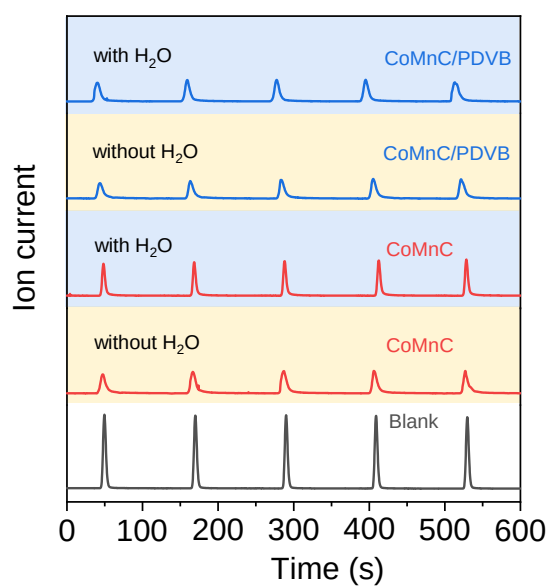


Figure S17. Transient response curves for pulsed CO (5 mL/min) in Ar flow (30 mL/min He) at 250 °C over the CoMnC and CoMnC/PDVB catalysts. Water bubbling with CO flow at 10 °C (water concentration at 1.2 vol% in CO) for introduction to the system.

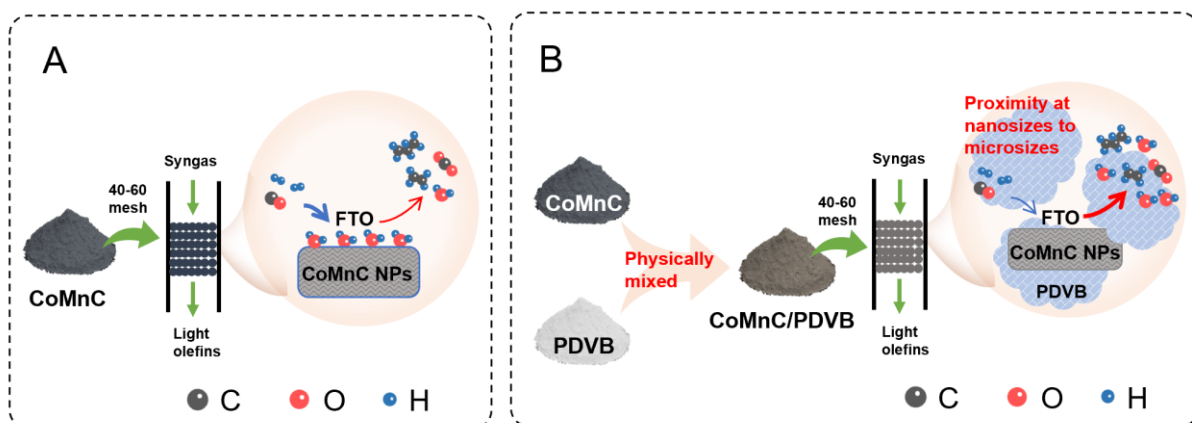


Figure S18. Scheme showing the CoMnC physically mixed with PDVB catalyst greatly boost the activity of syngas conversion *via* selectively and rapidly removing the water products from reaction system.

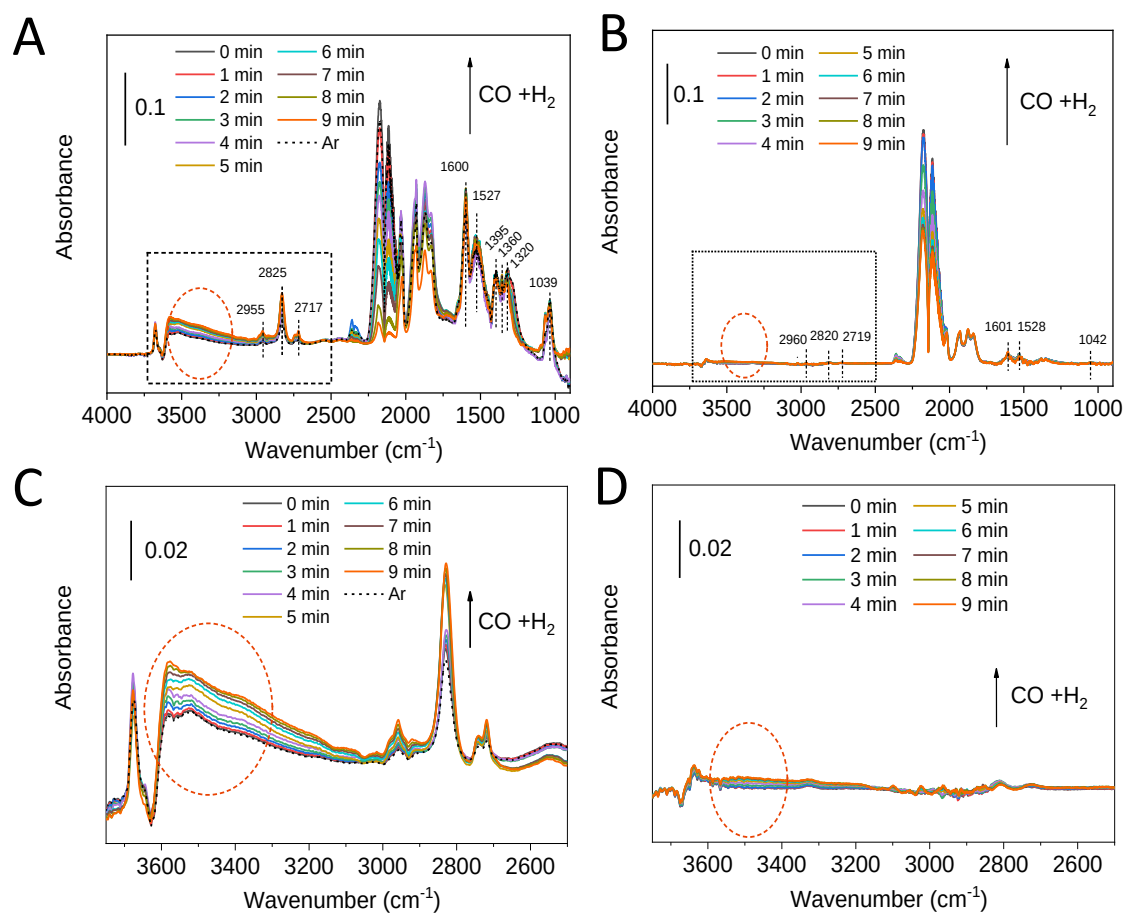


Figure S19. CO-hydrogenation *in-situ* FTIR spectra of the (A) CoMnC and (B) CoMnC/PDVB. (C) enlarge view of the region at 2500-3750 cm^{-1} in fig. S19A. (D) Enlarge view of the region at 2500-3750 cm^{-1} in fig. S19B.

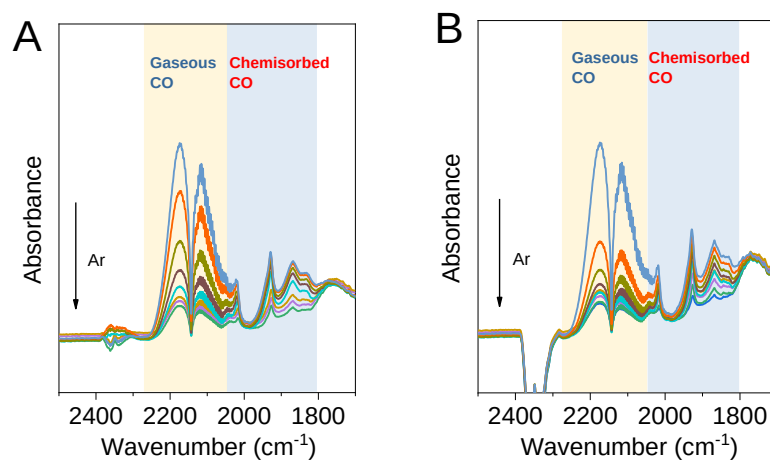


Figure S20. CO-desorption FTIR spectra of (A) anhydrous and (B) water-pretreated CoMnC/PDVB.

Note: Introducing CO to the CoMnC/PDVB catalysts showed the signals chemically adsorbed CO on the CoMnC surface in the *In-situ* FTIR spectra. By feeding water to the sample with CO gas, these signals were almost unchanged, supporting the negligible influence of water to CO adsorption on CoMnC, which is quite different from the water hindered CO adsorption on bare CoMnC without PDVB (Figs. 3D and E).

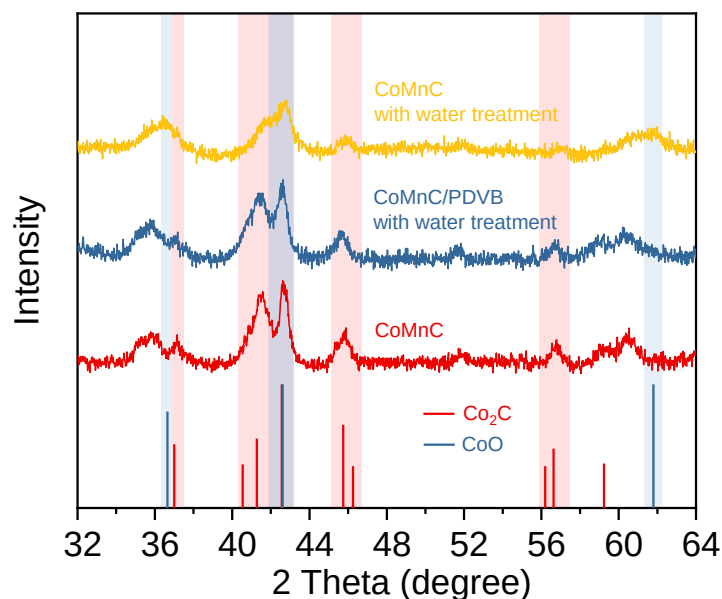


Figure S21. XRD patterns for studying the ability of the PDVB in improving the water resistance of CoMnC. The standard XRD patterns of the Co_2C and CoO is according to PDF # 65-1457 and PDF # 01-1227, respectively.

Note: The metal carbide catalysts for syngas conversion usually suffer from the oxidation by water to form the oxide phase under the long-period reaction or special treatment conditions. Therefore, we investigated the ability of the PDVB in improving the water resistance of CoMnC. Figure S21 showed that CoMnC could be oxidized by water treatment at 250 °C, as confirmed by the reduced Co_2C peaks and appearance of CoO peak in the XRD patterns. In contrast, the CoMnC with physically mixed PDVB was stable, giving almost unchanged XRD pattern before and after the treatment. This feature confirms that the PDVB could hinder the access of water to the cobalt carbide species to avoid the carbide oxidation, which helps to understand the hindered re-adsorption of water product to the CoMnC surface to benefit the CO adsorption in syngas conversion.

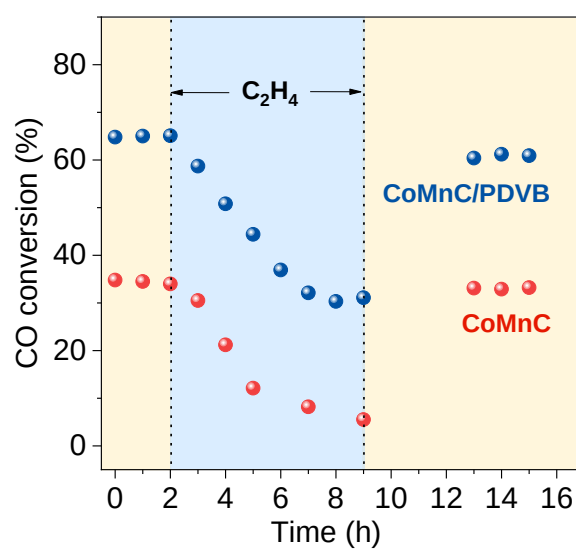


Figure S22. Data showing the influence of ethene to the syngas conversion on the CoMnC and CoMnC/PDVB catalysts. Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, ethene/H₂/CO at 0.6/2/1, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

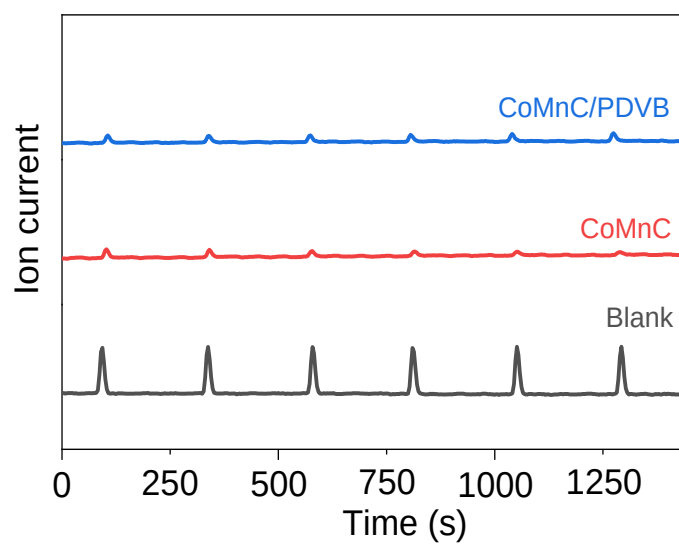


Figure S23. Transient response curves obtained during pulses of 10% CO/He (5 mL/min) into 10% C₃H₆ He flow (30mL/min He) at 250 °C over the CoMnC and CoMnC/PDVB catalysts. The CO signals (m/z at 28) were collected.

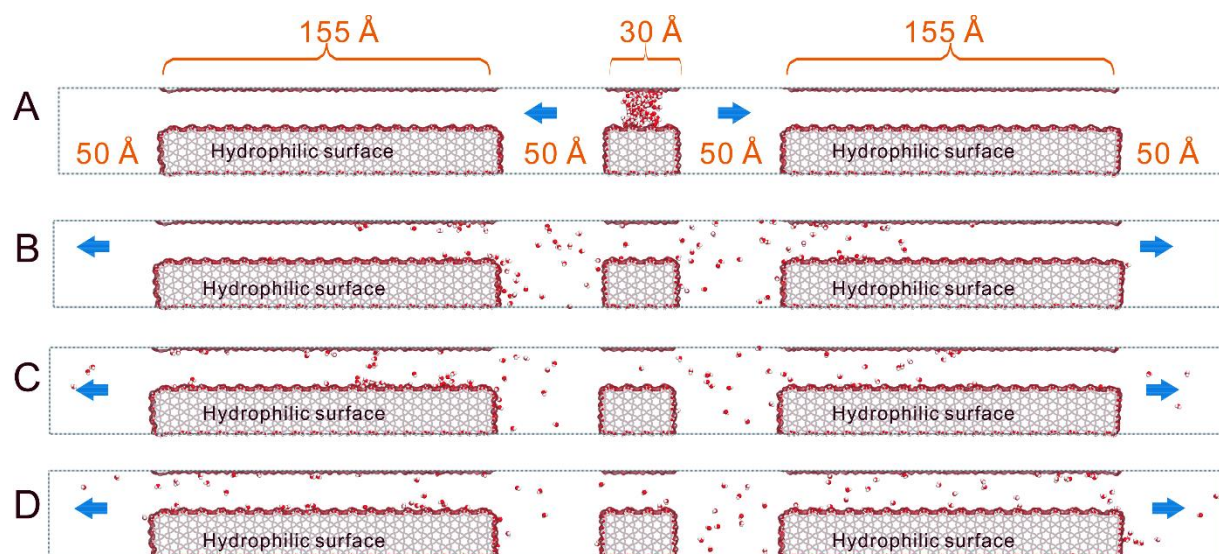


Figure S24. (A) The initial state of water molecules adsorbed on models with hydrophilic surface. Scheme showing the simulated diffusion of the water molecules on the models with hydrophilic surface at (B) 50, (C) 250, and (D) 500 ps.

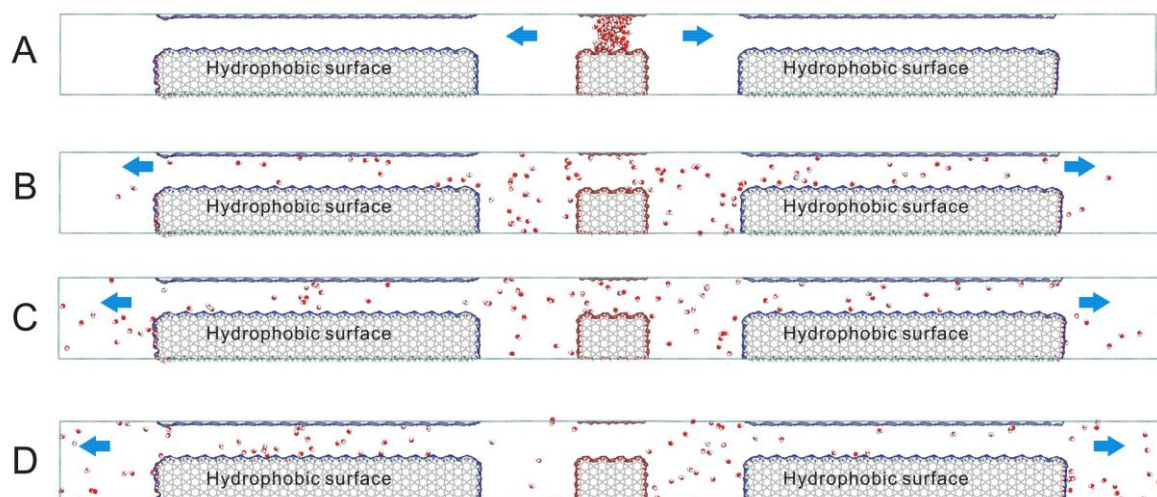


Figure S25. (A) The initial state of water molecules adsorbed on models with hydrophobic surface. Scheme showing the simulated diffusion of the water molecules on the models with hydrophobic surface at (B) 50, (C) 250, and (D) 500 ps.

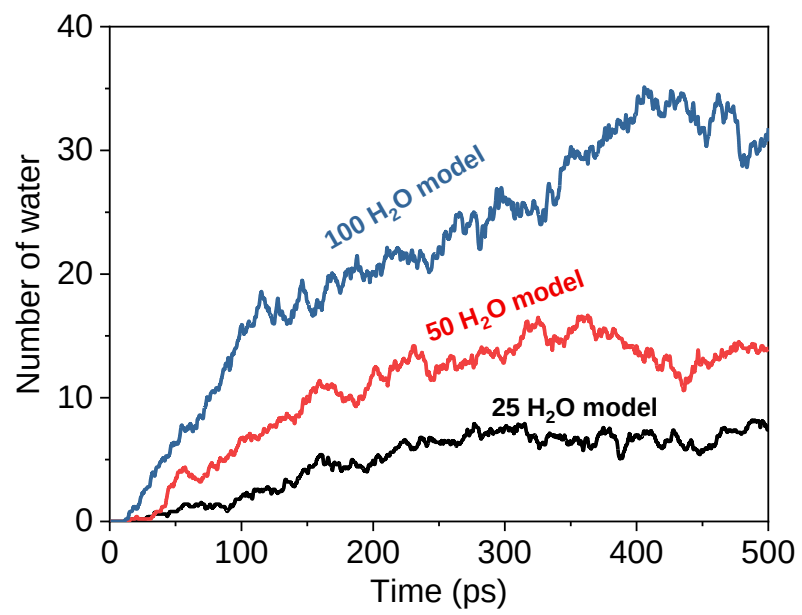
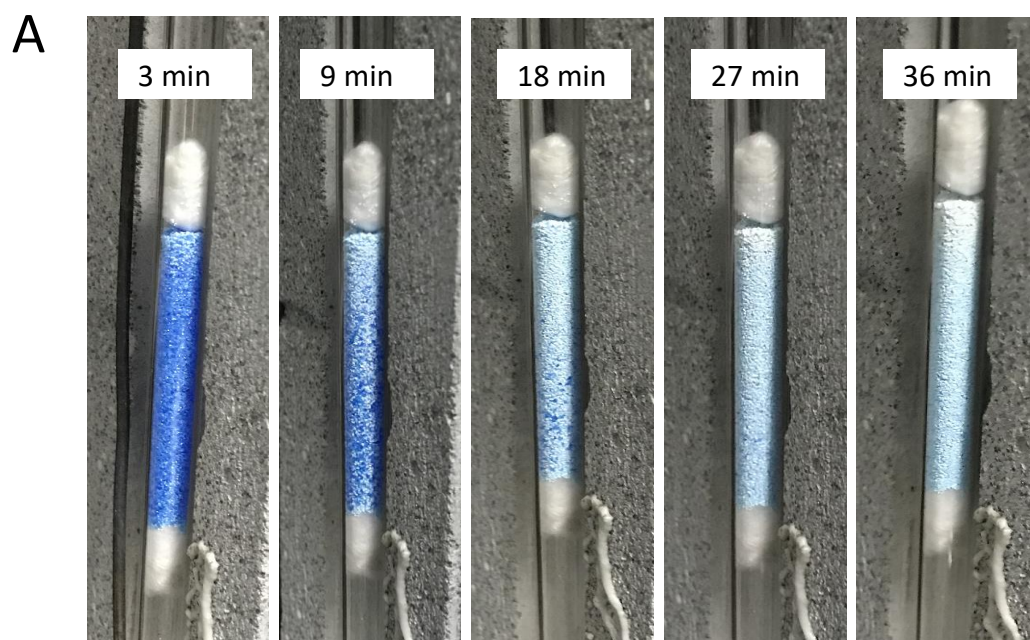
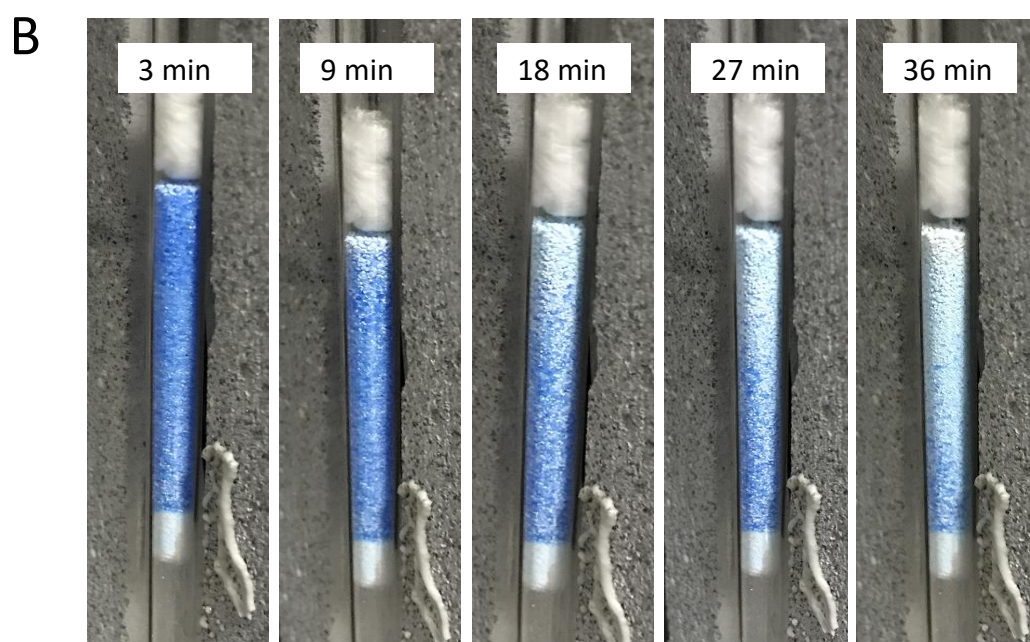


Figure S26. The number of water molecules escaped from hydrophobic model surface as a function of time.



2g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ + 15 mg PDVB



2g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Figure S27. Photographs of (A) 2 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ mixed with 15 mg of PDVB and (B) 2 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in a N_2 flow (20 mL/min) at 100 °C for different periods.

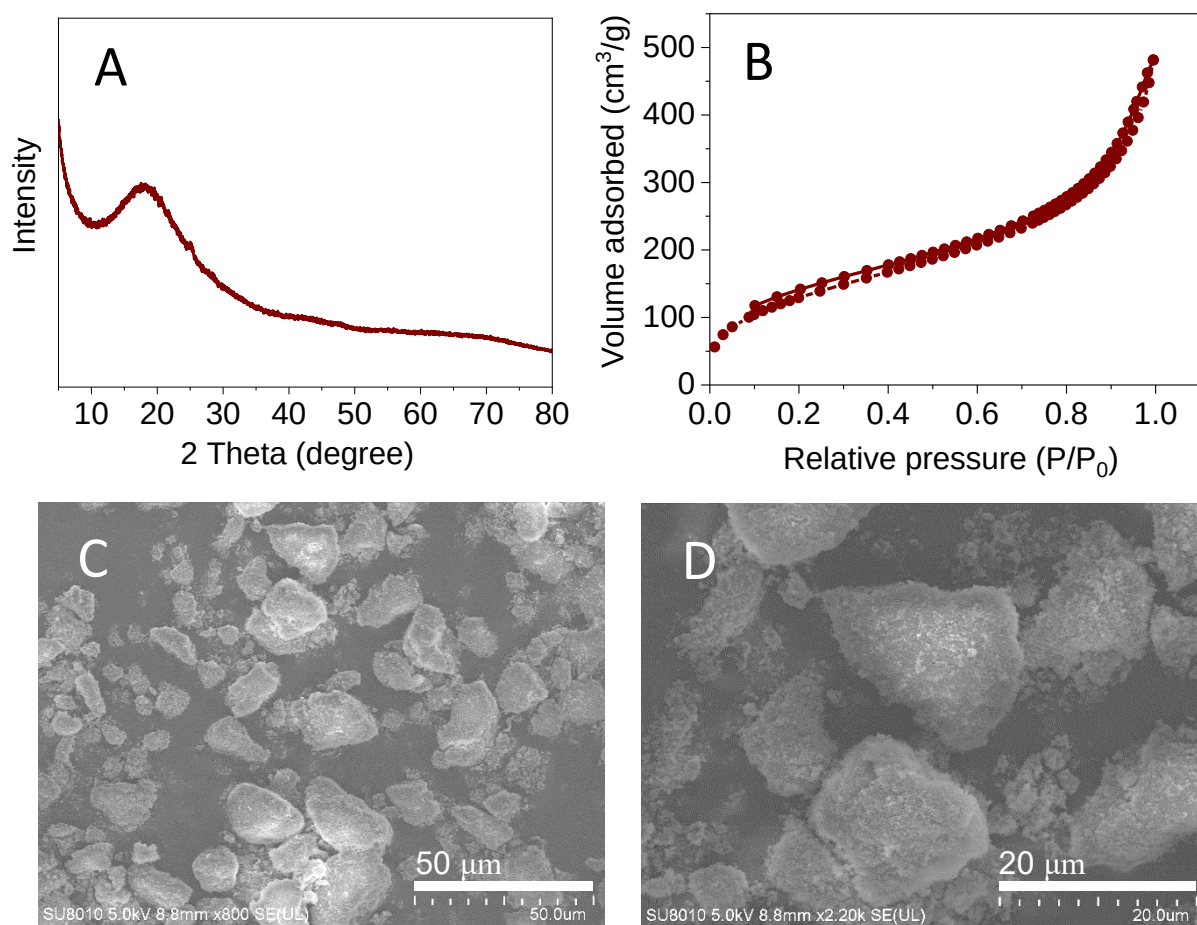


Figure S28. (A) XRD pattern, (B) N₂ sorption isotherms, and (C, D) SEM images of the PDVB-EA.

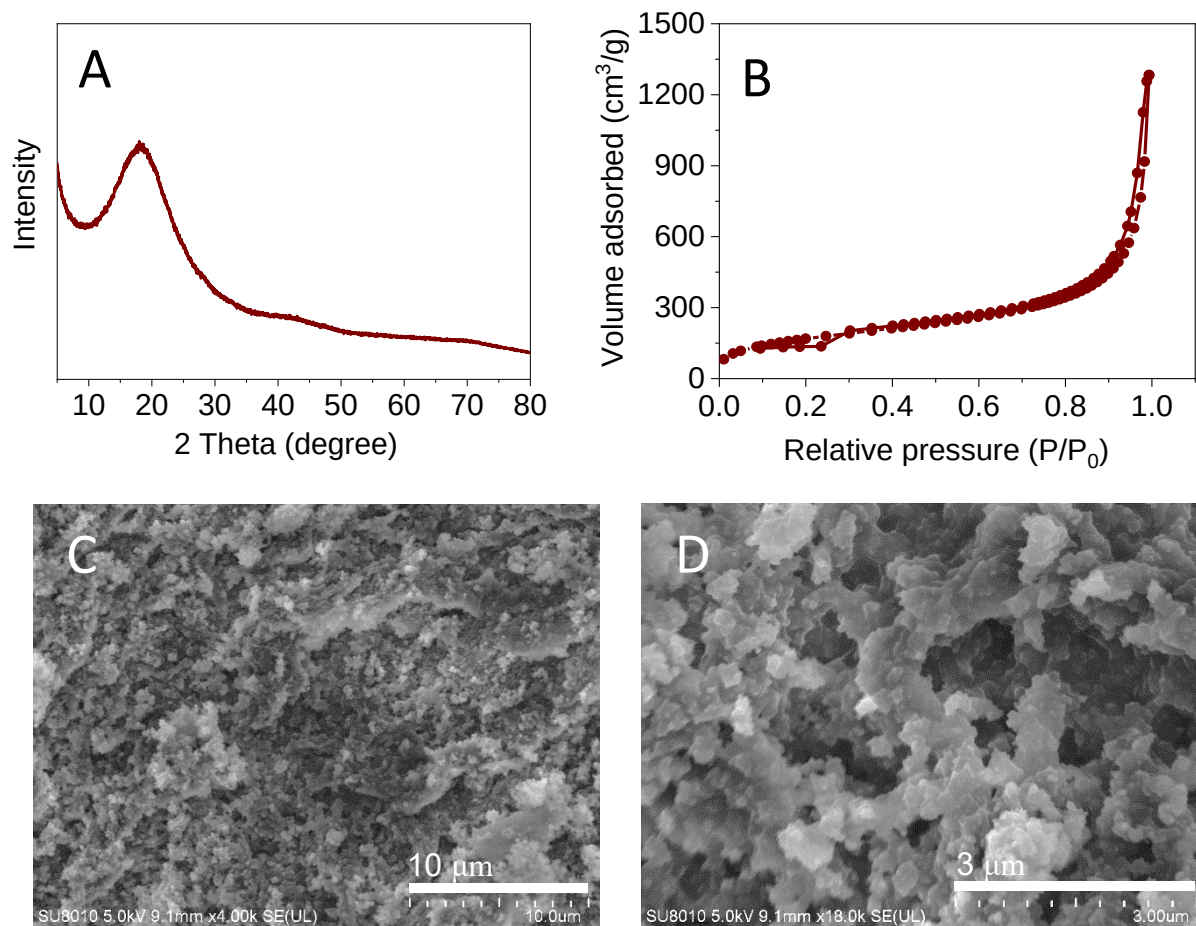


Figure S29. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the PDVB-DMF.

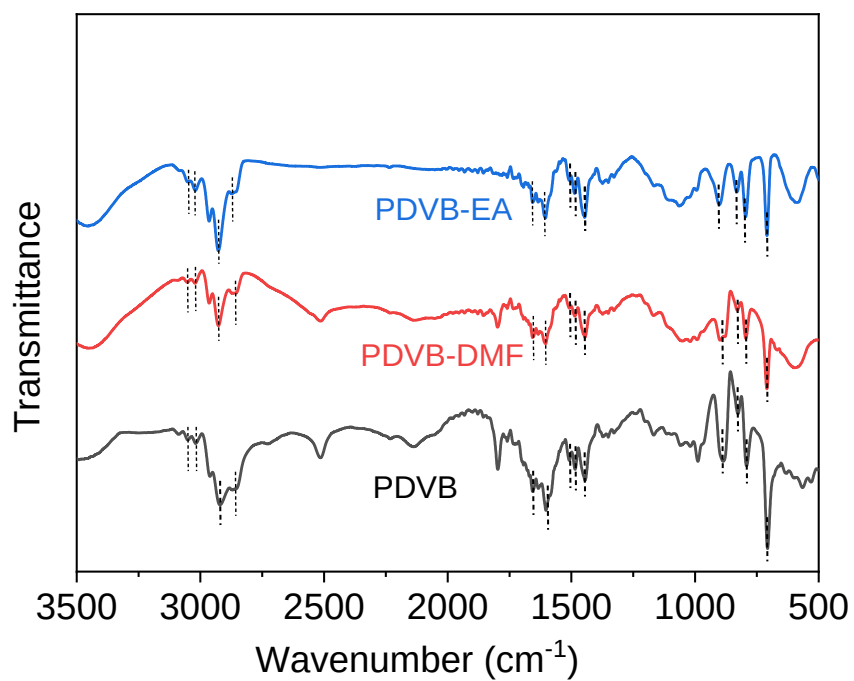


Figure S30. IR spectra of the PDVB, PDVB-DMF, and PDVB-EA.

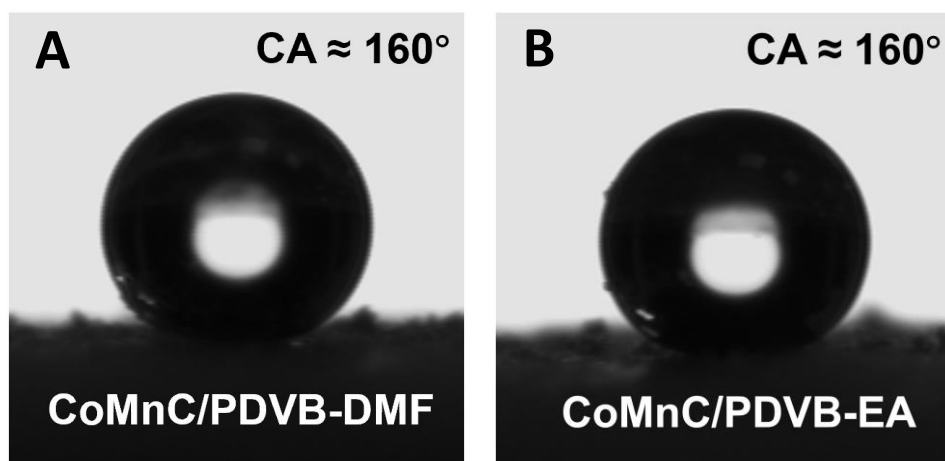


Figure S31. Water-droplet contact angles of the (A) CoMnC/PDVB-DMF and (B) CoMnC/PDVB-EA.

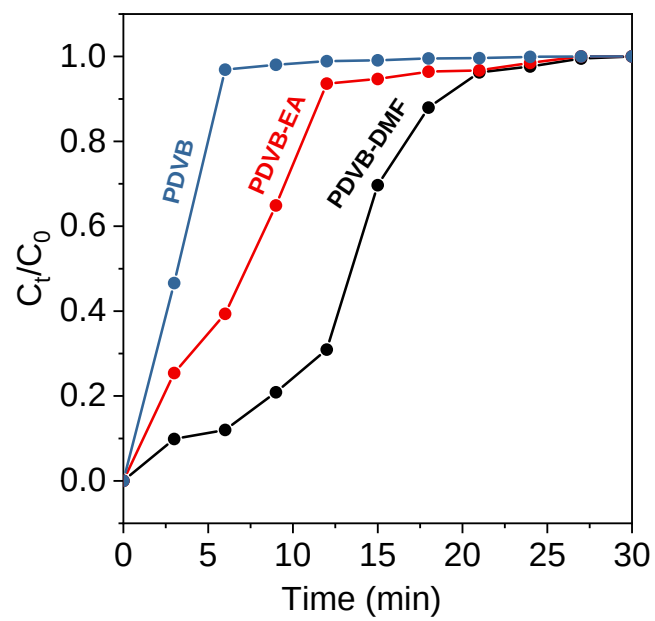


Figure S32. C_3H_6 breakthrough curves of the PDVB, PDVB-DMF, and PDVB-EA.

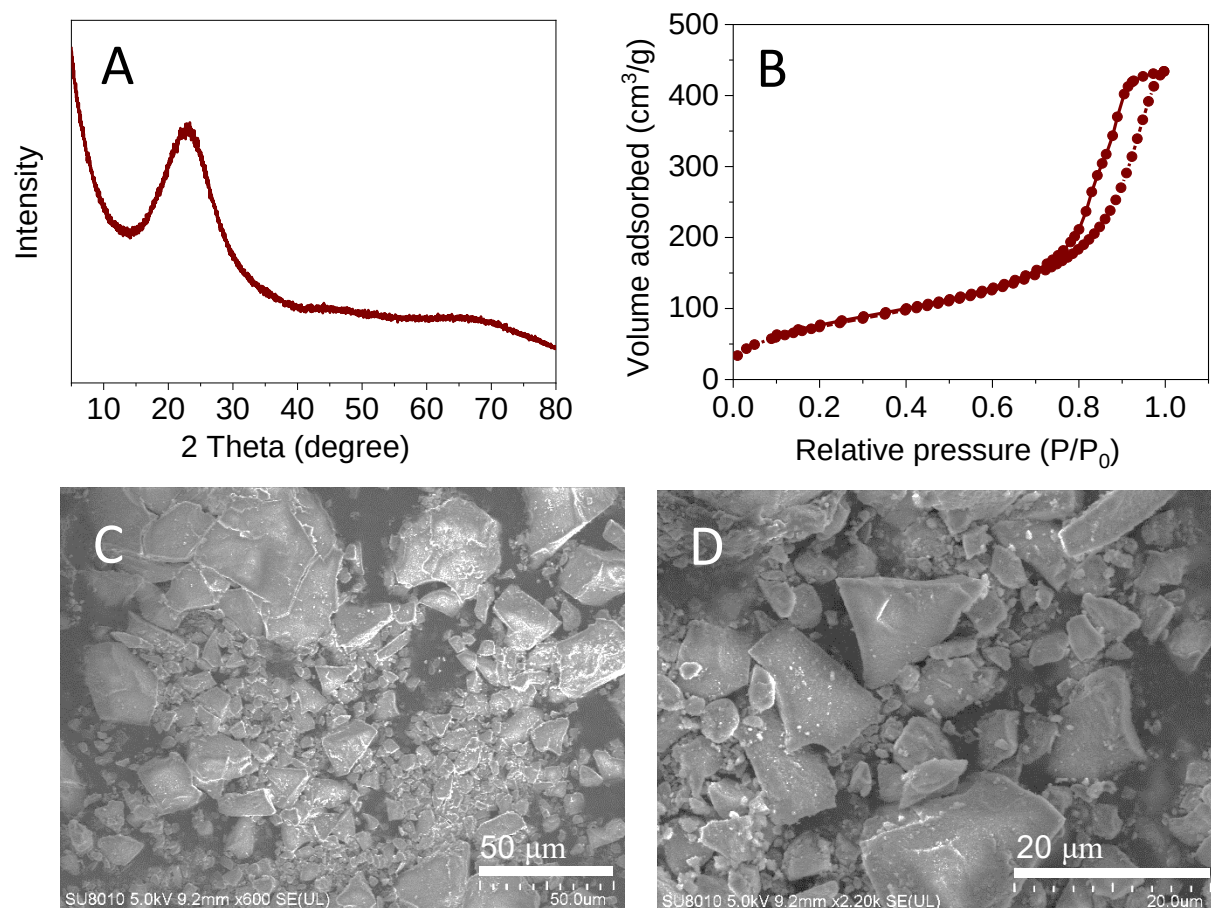


Figure S33. (A) XRD pattern, (B) N₂ sorption isotherms, and (C,D) SEM images of the SiO₂-Me.

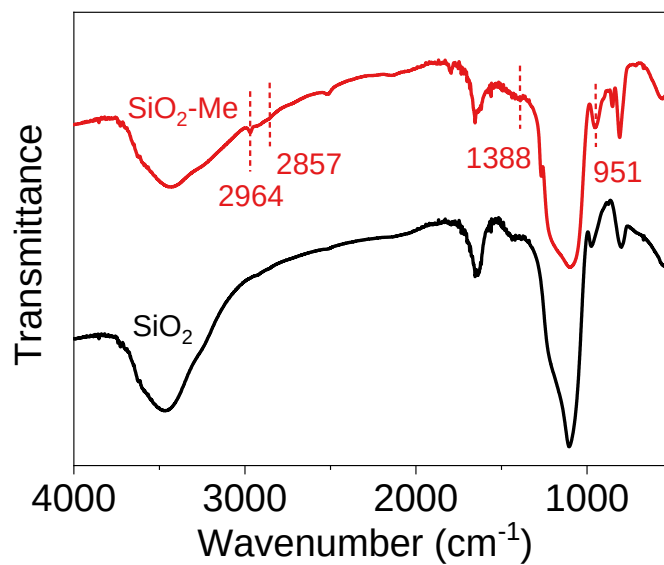


Figure S34. FTIR spectra of SiO₂ and SiO₂-Me.

Note: Compared with SiO₂, the SiO₂-Me gave additional peaks at 2964, 2857, 1388, and 951 cm⁻¹, which were assigned to the asymmetrical stretching vibration of -CH₃ (2964 cm⁻¹), symmetrical stretching vibration of -CH₃ (2857 cm⁻¹), and bending vibration of -CH₃ (1394 cm⁻¹), confirming the successful modification of methyl groups on SiO₂ (57).

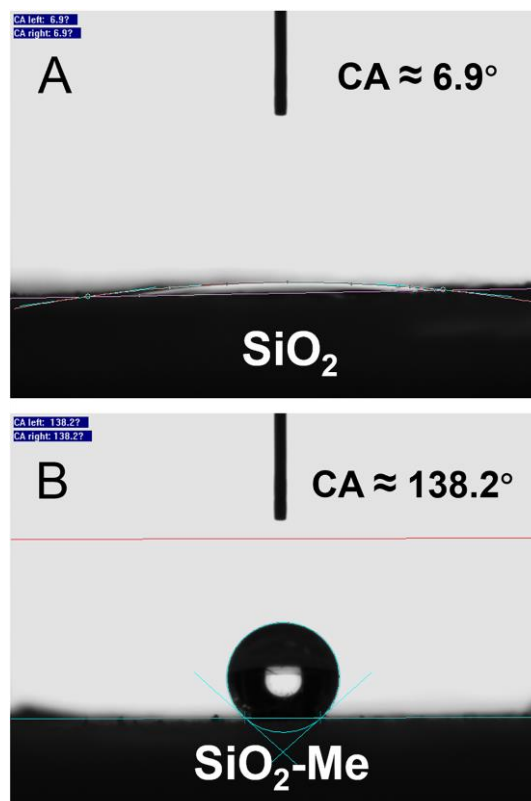


Figure S35. Water-droplet contact angles of the (A) SiO_2 and (B) $\text{SiO}_2\text{-Me}$.

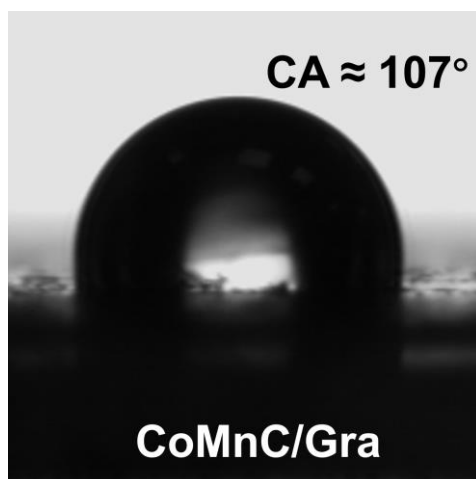


Figure S36. Water-droplet contact angle of the CoMnC/Gra catalyst.

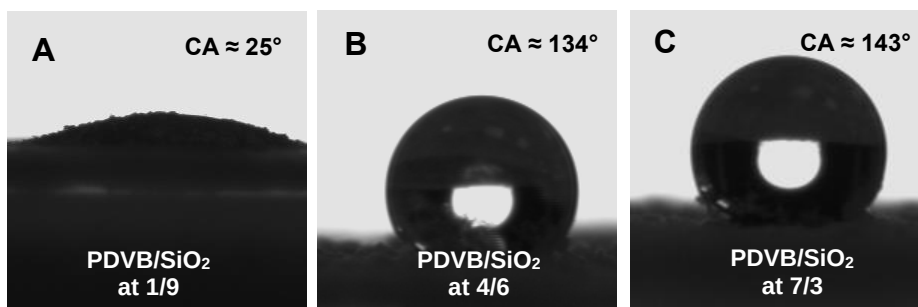


Figure S37. Water-droplet contact angles of the PDVB and SiO₂ mixtures with different PDVB/SiO₂ weight ratio at (A) 1/9, (B) 4/6, and (C) 7/3, respectively.

Note: The materials with different wettability were mixed with CoMnC and evaluated in the syngas conversion. We reasonably mixed hydrophobic PDVB and hydrophilic silica with different ratios, and the wettability of the mixtures could be varied and then used as promoters for the CoMnC catalyzed syngas conversion. Fig. S37 shows water-droplet contact angles of 25, 134, and 143° for the mixtures with different PDVB/SiO₂ weight ratios at 1/9, 4/6, and 7/3, respectively. In the syngas conversion, the CO conversions were positively related to the hydrophobicity (table S8). The conversion was 34.8% on the CoMnC catalyst containing the hydrophilic promoter (PDVB/SiO₂ weight ratio at 1/9), and increased to 52.4% and 65.9% when relatively hydrophobic PDVB/SiO₂ mixtures (PDVB/SiO₂ weight ratio at 4/6 and 7/3) were employed. In contrast, the PDVB without SiO₂ gave a CO conversion at 72.2%, supporting the importance of promoter hydrophobicity for the catalysis. In this case, more hydrophobic promoter also slightly reduced the olefin selectivity as confirmed by the changed o/p ratio in these reactions, because the rapid water escape from the catalyst surface could lead to the re-adsorption and deep hydrogenation of olefin products.

We have to note that the CoMnC and PDVB/SiO₂ mixed catalysts were made into granule mixture with sizes at 80-100 mesh in the aforementioned tests rather than the powder mixture to avoid the interaction between silanol groups with the CoMnC surface during the reaction. According to the catalysts over CoMnC/PDVB catalysts with different mixing manners in power and 80-100 mesh granules, the catalytic performances are comparable in these cases.

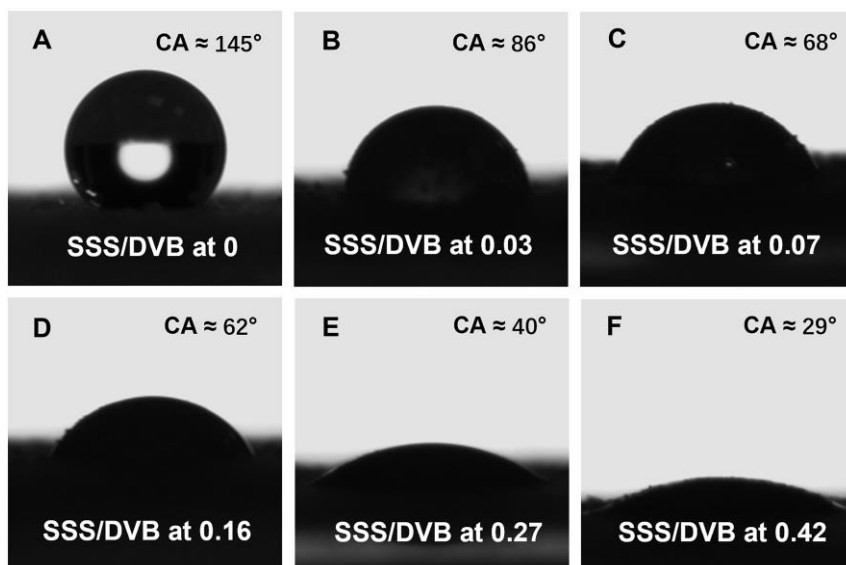


Figure S38. Water-droplet contact angles of the PDVB-SSS materials with varied molar ratios of SSS and DVB in the starting mixture at (A) 0, (B) 0.03, (C) 0.07, (D) 0.16, (E) 0.27, and (F) 0.42.

Note: We changed the wettability of the polymer support from hydrophobic to relatively hydrophilic by introducing the hydrophilic sodium styrene sulfonate (SSS) to the PDVB. By varying the molar ratios of SSS to DVB in the starting mixture to 0.03, 0.07, 0.16, 0.27, and 0.42, the PDVB-SSS materials with water-droplet contact angles at 86, 68, 62, 40, and 29° were obtained, respectively (fig. S38). As shown in table S9, more SSS in the PDVB-SSS materials have lower CO conversion, showing the CO conversion at 34.9, 28.8, and 25.2% over the PDVB-SSS with SSS/DVB at 0.03, 0.07, and 0.16, respectively. For more hydrophilic samples of PDVB-SSS with SSS/DVB at 0.27 and 0.42, the CO conversions at 5.9-6.2% were obtained, supporting the importance of hydrophobic promoter.

These data confirm that the hydrophobic PDVB is crucial for realizing the desired performances, and addition of hydrophilic individual to the promoter obviously reduced the activity and/or selectivity in CoMnC catalyzed syngas conversion. However, it is also notable that the promoter with different compositions might also lead to other unknown factors to influence the catalysis, therefore the effect of wettability on catalytic performances can only be compared within the promoters with the same components (e.g. PDVB/SiO₂ or PDVB-SSS).

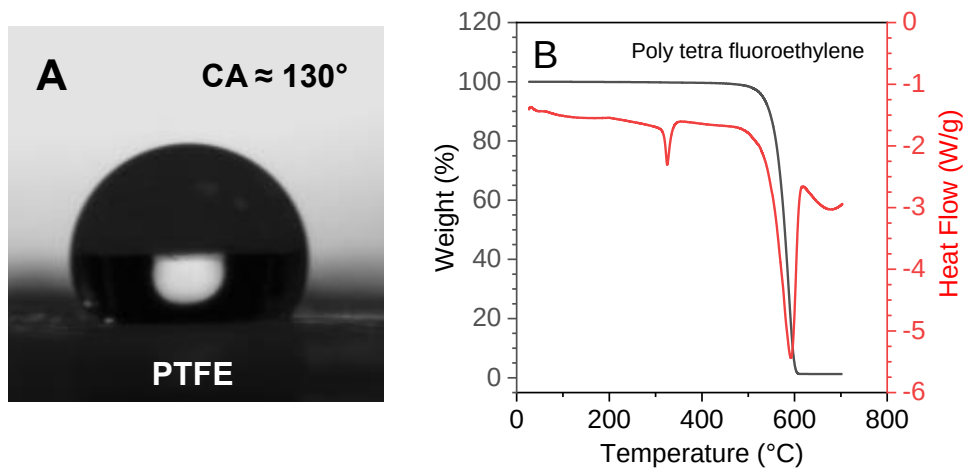


Figure S39. (A) Water-droplet contact angle and (B) TG/DSC profiles of PTFE.

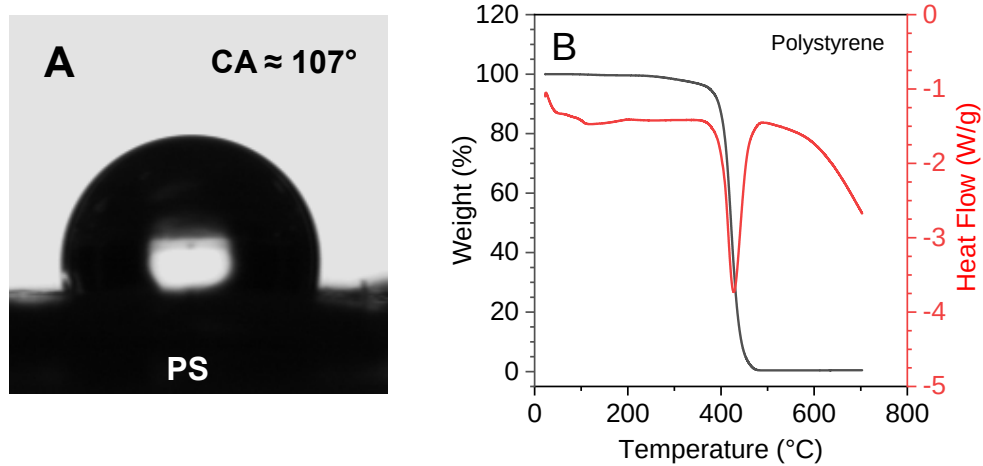


Figure S40. (A) Water-droplet contact angle and (B) TG/DSC profiles of PS.

Table S1. Catalytic data of various catalysts in syngas conversion.^[a]

Catalyst	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
CoMnC	32.2	48.1	3.3	2.4	60.8	33.5	15	36	30	25
CoMnC/PDVB	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
CoMnC ^[b]	15.4	44.7	9.7	6.0	72.3	12.0	6	24	21	12
CoMnC/PDVB ^[b]	50.7	47.4	6.1	4.2	73.8	15.9	9	31	24	18
CoMnC ^[c]	5.5	42.8	15.7	5.5	70.7	8.1	8	29	19	13
CoMnC/PDVB ^[c]	19.4	46.2	10.9	4.2	72.2	12.7	12	26	19	17
CoMnC/PDVB-EA	88.7	49.3	4.2	8.7	53.4	33.7	1	12	11	6
CoMnC/PDVB-DMF	92.4	48.5	5.3	13.4	37.6	43.7	0.4	4	4	3
CoMnC/SiO ₂ -Me	74.7	49.2	7.6	15.5	45.5	31.4	1	7	6	3
CoMnC/Gra	53.0	47.5	6.8	4.5	67.3	21.4	7	26	19	15
Used-CoMnC ^[d]	26.2	48.4	7.7	5.9	54.9	31.5	4	16	16	9
CoMn@Si	23.7	4.3	27.9	5.8	25.9	40.4	1	9	8	4
CoMn@Si-c	22.2	4.5	28.9	6.2	26.5	38.4	1	9	7	4

[a] Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of promoter, or 1.0 g of silica modified CoMn catalysts, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoters.

[b] 3600 mL·h⁻¹·g⁻¹, 270 °C.

[c] 7200 mL·h⁻¹·g⁻¹, 270 °C.

[d] The used-CoMnC was obtained by separating the PDVB component of the used CoMnC/PDVB catalyst and reused in the next run.

Table S2. Surface area and pore volume of various samples by N₂ sorption measurement.

Catalyst	S _{BET} (m ² g ⁻¹)	V (cm ³ g ⁻¹)
SiO ₂ -Me	278.2	0.67
PDVB-EA	488.2	0.74
PDVB-DMF	623.3	1.98
PDVB	<5	-

Table S3. Distributions of hydrocarbon products in syngas conversion over the CoMnC and CoMnC/PDVB catalysts.^[a]

Catalyst	Selectivity (C%, CO ₂ -free)													C ₈ & C ₈₊
	CH ₄	C ₂ ⁰	C ₂ ⁼	C ₃ ⁰	C ₃ ⁼	C ₄ ⁰	C ₄ ⁼	C ₅ ⁰	C ₅ ⁼	C ₆ ⁰	C ₆ ⁼	C ₇ ⁰	C ₇ ⁼	
CoMnC	3.3	0.9	12.9	0.9	30.8	0.6	17.1	0.5	12.2	0.3	7.5	0.2	5.1	7.7
CoMnC/P DVB	4.8	4.1	13.0	1.9	37.4	0.9	21.0	0.3	8.5	0.3	4.4	0.1	1.7	1.6

[a] Reaction conditions: 1.0 g of CoMnC or 1.0 g of CoMnC physically mixed with 1.0 g of promoter, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and promoter.

Table S4. Catalytic data of the reaction with different feed gas over the CoMnC/PDVB catalyst.^[a]

H ₂ /CO/CO ₂ /Ar	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
2	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
3	80.2	41.3	12.4	4.6	54.8	28.2	5	21	17	12
4	88.7	38.7	15.8	6.0	49.6	28.6	3	17	14	8
3.6 ^[b]	61.7	23.7	16.7	7.0	59.5	16.8	3	21	17	8

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of PDVB, H₂/CO at 2, GHSV at 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

[b] The component feed gas is H₂/CO/CO₂/Ar at 68/19/10/3.

Note: We optimized the molar ratio of H₂/CO in the feed gas, reducing the CO₂ selectivity to 38.7% with H₂/CO feed ratio as high as 4. Adjusting the H₂/CO ratio to 3.6 and introducing a small amount of CO₂ in the syngas feed could remarkably reduce the CO₂ selectivity in the products to 23.7%. These data confirm that CO₂ in the feed is necessary for further reduction of the CO₂ selectivity in the products.

Table S5. Data showing the performances of various catalysts reported in literature.

Catalyst	T (°C)	P (MPa)	H ₂ /CO (molar)	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (C%, CO ₂ -free)				Yield of C ₂ -C ₄ olefins (%)	Ref.
						CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊		
CoMnC/PDVB	250	0.1	2	63.7	49.0	4.7	6.8	71.2	17.3	23.1	TW ^[a]
CoMn	250	0.1	2	31.8	47.3	5.0	2.0	60.8	32.2	10.2	8
FeN-Mn/CNT	300	0.5	1	11.8	36.1	23.6	8.3	43.9	24.2	3.3	42
6wt%Mn/Fe ₃ O ₄	320	1.0	1	41.5	37.8	9.7	6.5	60.1	23.7	15.5	43
Fe-Na-S/ α -Al ₂ O ₃	340	2.0	1	80.0	50.0	11.0	6.0	53.0	30.0	21.2	14
Fe/N-Gra	340	0.5	1	21.1	35.1	21.4	9.2	49.6	19.8	6.8	44
ZnCrO _x /MSAPO-34	400	2.5	2.5	17.0	45.0	2	14.0	80.0	4.0	7.5	15
ZnO-ZrO ₂ /SAPO-34	400	1.0	2	11.0	45.0	4.0	18.0	74.0	4.0	4.5	16
ZnO-ZrO ₂ /SSZ-13	400	3.0	2	29.0	45.0	2	16.0	77.0	4.0	12.3	45
ZrO ₂ -In ₂ O ₃ /SAPO-34	400	2.0	1	13.1	40.0	6.2	12.5	79.9	1.3	6.3	46

[a] This work.

Table S6. Catalytic data of catalysts with different mixing manners of the CoMnC and PDVB.^[a]

Mixture manner	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
Dual-bed	29.1	46.3	7.8	2.4	61.2	28.6	16	34	27	25
10-20 mesh	47.7	46.6	7.5	7.8	58.6	26.1	3	13	15	7
20-40 mesh	52.2	46.9	7.0	7.6	59.8	25.6	3	15	15	8
40-60 mesh	53.7	46.8	6.8	6.0	66.4	20.8	4	18	18	11
Powder-mixing	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10

[a] Reaction conditions: 1.0 g of CoMnC and 1.0 g of PDVB, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C.

Table S7. Catalytic data of the CoMnC/PDVB catalysts with different PDVB/CoMnC weight ratios of 0.5, 1, and 1.5.^[a]

PDVB/CoMnC (weight ratios)	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
0.5	56.4	48.2	9.8	3.0	62.8	24.4	10	31	29	21
1	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
1.5	70.5	48.5	8.9	3.3	64.5	23.3	8	29	25	20

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 0.5, 1.0 and 1.5g of PDVB, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, powder-mixing for CoMnC and PDVB.

Table S8. Data showing the performances of CoMnC catalyst mixed with PDVB/SiO₂ promoters in the syngas conversion.^[a]

Weight ratios of PDVB/SiO ₂ in the promoter	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
1/9	34.8	48.6	4.5	3.0	62.7	29.8	11	30	24	21
4/6	52.4	48.0	4.6	3.3	65.4	26.7	10	30	23	20
7/3	65.9	46.9	4.8	3.5	63.2	28.5	8	27	22	18

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of promoter (80-100 mesh), H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, and 250 °C.

Table S9. Data showing the performances of CoMnC catalyst mixed with PDVB-SSS promoters with different molar ratios of SSS/DVB.^[a]

Molar ratio of SSS/DVB	CO Conv. (%)	CO ₂ Sel. (%)	Selectivity (C%, CO ₂ -free)				o/p			
			CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C-oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
0	63.5	46.3	4.8	6.9	71.4	16.9	3	20	25	10
0.03	34.9	45.9	5.0	2.1	57.2	35.7	17	35	27	27
0.07	28.8	44.4	5.5	2.1	59.2	33.2	20	36	27	29
0.16	25.2	45.9	6.1	2.6	56.5	34.8	12	32	26	22
0.27	5.9	46.7	8.0	2.2	67.2	22.6	21	40	34	30
0.42	6.2	40.3	7.9	2.4	64.6	25.1	18	38	32	27

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of PDVB modified by different amounts of sulfonate, H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, 250 °C, and powder-mixing.

Table S10. Data characterizing the performances of the CoMnC catalyst mixed with different hydrophobic materials in FTO reaction.^[a]

Catalyst	CO	CO ₂	Selectivity (C%, CO ₂ -free)				o/p			
	Conv. (%)	Sel. (%)	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊ & C- oxy.	C ₂	C ₃	C ₄	C ₂ -C ₄
CoMnC/PTFE	37.4	44.4	12.5	4.1	59.7	23.7	6	27	24	14
CoMnC/PS	10.7	46.7	37.8	10.0	37.8	14.4	2	7	9	4

[a] Reaction conditions: 1.0 g of CoMnC physically mixed with 1.0 g of promoter catalysts (40-60 mesh), H₂/CO at 2, 1800 mL·h⁻¹·g⁻¹, 0.1 MPa, and 250 °C.

References and Notes

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